YILMAZLAR BIOGAS POWER PLANT PROJECT

Document Prepared by Earthood Services Private Limited

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## Summary:

Earthood Services Private Limited (ESPL) has been contracted by Yılmazlar Madencilik Sanayi ve Ticaret Limited Şirketi to carry out the joint validation and verification of their VCS project " Yilmazlar biogas power plant project" with project ID 4205 against VCS Standard v4.4/11/.

The Project undertakes renewable energy generation through biomethanization systems, the biogas to energy systems captures biogas from animal manure via anaeerobic digestion and utilizes it to produce thermal and electric energy through cogeneration systems. The project enables reduction of GHG incurred from existing system of cattle manure generated at farms, which is left to decay at and around farms in anaerobic conditions.

The Biogas power plant is located in Derbent Village of Tavşanlı District of Kütahya Province in Türkiye. The total area of the project is 24.330.18 $m^2$. Derbent village is 1250 meters north of the project, Yeniköy village is 2120 meters south, and Demirbilek Village is 3500 meters west which are the nearest settlements.

The purpose of the joint validation and verification of the project is to have a thorough and independent assessment of the proposed Project against the following –

- Applicable VCS requirements, in particular, the project's baseline, monitoring plan and the compliance with relevant VCS rules and Host Party criteria.
- It includes confirming the implementation of the monitoring plan in the current monitoring.
- The application of methodologies, AMS-III.D., "Methane Recovery in animal manure management systems", version 21.0 and AMS-I.D., "Grid-connected renewable electricity generation "version 18.0 /7A, 7B/.

  These are validated and verified to confirm that the project design, as documented, is sound and reasonable and meets the identified criteria and assess the claims and assumptions made without limitation on the information provided by the project proponents.

The scope of the services provided by the Earthood Services Private Limited is to conduct an independent and objective assessment of the Project Description (PD) to compare the assertions and presumptions in the project description and ER sheet /40/ to the VCS criteria, including but not limited to the VCS Standard, other pertinent guidelines established for Verra, UNFCCC criteria and the applied methodology, AMS-III.D, version 21.0 and AMS-I.D version 18 .0 /7A, 7B/.

The joint validation and verification of the Project was undertaken in the following manner -

- A desk review of the project description and supporting documents.
- A physical site visit and interviews with the stakeholders
- Resolution of outstanding issues
- Issuance of final Joint validation and verification report

During validation, a total of 11 findings were raised, which includes 07 Corrective Action Request, 04 Clarification Request, 00 Forward Action Request. All the findings have been successfully addressed by the Project Proponent and subsequently closed by the validation team; the details of which are provided under Appendix IV of this report.

The review of the joint PD & MR, supporting documentation and subsequent follow up actions have provided Earthood with sufficient evidence to determine the fulfilment of the stated criteria. There are no uncertainties associated with the validation of the Project and a reasonable level of assurance has been obtained.

To summarize, Earthood Services Private Limited is of the opinion that the Project "Yilmazlar Biogas Power Plant", VCS ID: 4205 "as described in the PD & MR, version 4.0 dated 12/01/2024/1/ meets all applicable VCS requirements, including those specified in the VCS Standard version 4.4/4/, relevant methodology, tools and guidelines. The selected baseline and monitoring methodology, AMS-III.D and AMS-I.D /7A, 7B/ is applicable to the Project and have been applied appropriately.

The project is estimated to generate 11,700 MWh/year of electricity /22/ and is expected to reduce a total of 35,855 $tC0_2e$ over the renewable crediting period of 7 years (29/09/2021 to 28/09/2028). The Project in the current monitoring period from 29-September-2021 – 31-March-2023 has reduced a total of 37,150 $tCO_2e$ and has supplied 8,989 MWh of electricity to the grid displacing the fossil fuel dominated grid electricity. Therefore, Earthood requests the registration and issuance of the project as a VCS Project.

# CONTENTS

# 1 INTRODUCTION

## 1.1 Objective

Yılmazlar Madencilik Sanayi ve Ticaret Limited Şirketi (PP) has contracted Earthood Services Private Limited (VVB) for the VCS project "Yılmazlar Biogas Power Plant Project" (project ID 4205) located in India to jointly carry out validation and verification of the project against the requirement of VCS Program.

The objective of the validation and verification assessment is to undertake an independent third-party assessment and determine the

- Project conformance to VCS rules;
- Project conformance to the applied methodology, including the procedure for the demonstration of additionality specified in the methodology; and
- Likelihood that methods and procedures set out in the project description will generate verifiable GHG data and information when implemented.

In order to fulfil the above objectives, the VVB assigned the task of assessment to validation/ verification team that is collectively qualified as per Annex A.2.3.3 of ISO 14065 and other internal requirements.

The validation/ verification is an independent assessment of the project by a validation/verification body that determines whether the project complies with the VCS rules.

## 1.2 Scope and Criteria

The scope of the validation is defined as an independent and objective review of the VCS Project Description and Monitoring Report /1/ and supporting documents, which the assessment team reviewed against the pertinent criteria and decisions by the VCS Board, including but not limited to, the CDM methodologies/7A, 7B/, applicable tools/8/, VCS Standard/4/and VCS Program Guide/3/. Based on the recommendations mentioned in the latest version of VCS Validation and Verification Manual, version 3.2/5/, the assessment team has employed a risk-based approach, focusing on identifying the significant risks for implementation of the project and generating VCUs. The scope of this assignment is:

- Validation in accordance with Earthood's own QMS that is based on the VCS VVM along with the guidance provided by the VCS Board to determine if the project meets all the relevant VCS requirements, including those specified in the VCS Standard/4/, relevant methodology/7A, 7B/, tools/8/ and guidelines and processing in line with the VCS Program Guide/3/.

- To assess the accuracy, conservativeness, relevance, completeness, consistency, and transparency of the information provided by the Project Proponent.
- To report the findings and conclusions in an objective manner and conduct the validation in accordance with VCS Rules and procedures.
- To apply consistent validation criteria in providing expert judgements to the requirements of applicable approved methodologies, tools, and cross check the same.
- Safeguard the confidentiality of all information obtained during validation; and to adhere to the principles of independence, ethical conduct, fair presentation, and due professional care in assessment process.

## 1.3 Reasonableness of Assumptions and Level of Assurance

☒ Reasonable Level of Assurance

☐ Limited Level of Assurance

The level of assurance of the joint validation and verification report falls under reasonable assurance engagements. Reasonable assurance is high level of assurance regarding material misstatement, but not an absolute one.

Reasonable assurance includes the understanding that there is a remote likelihood that material misstatement will not be prevented or detected on a timely basis. To achieve reasonable assurance, the auditor needs to obtain sufficient appropriate audit evidence to reduce audit risk to an acceptable low level. This means that there is some uncertainty arising from the use of sampling, since it is possible that a material misstatement will be missed.

The evidence used to achieve a reasonable level of assurance is specified in section 2.3 and 2.4 of this report.

## 1.4 Summary Description of the Project

The Project undertakes development of Yilmazlar Biogas Power Plant, a biogas to energy project that will generate renewable energy by capturing biogas from cattle manure -via anaerobic digestion- and utilizing it to produce thermal and electric energy through cogeneration systems. The project is being implemented by Yılmazlar Madencilik Sanayi ve Ticaret Limited Şirketi which has constructed a cattle farm called Kayi farm in Derbent village. This farm has been operational since year 2013, with respect to the manure generated from the farm activities, the project proponent has invested in construction of the BPP, and it turned operational from June 2021.

The purpose of the project is to enable reduction of GHG incurred from existing system of cattle manure generated at farms, which is left to decay at and around farms in anaerobic conditions. The waste from neighbouring Kayi cattle farms and 11 poultry farms/21/ will be collected and fed into anaerobic digesters at the biogas power plant.

The Project proposes installation of 1 gas engine unit thereby reaching an installed operation capacity of 1.560 Mwe, commissioned on 27/06/2021 /41/were found operational. The generation system at full operational capacity is estimated to generate 11,700 MWh of annual feed-in electricity/22/

Whilst providing sustainable development benefits to the host communities and the host country, the proposed Project will reduce greenhouse gas (GHG) emissions by:

- Preventing GHG emissions, methane, from being emitted directly to the atmosphere from cattle manure that would be otherwise left to decay in aerobic conditions.
- Replacing the electricity that would have otherwise been generated by the national grid which is heavily dependent on fossil-fuel-based resources, through generating renewable energy and feeding it to the grid.
- Producing organic compost from solid waste stored in anaerobic digesters.

The Project is estimated to offset an annual average of 250,983 $tC0_2e$ emission resulting in a total of 35,855 $tC0_2e$ emission reductions over the renewable crediting period of 7 years (29/09/2021 to 28/09/2028).

# 2 VALIDATION AND VERIFICATION PROCESS

## 2.1 Method and Criteria

The validation and verification of the project consists of the following steps –

- Desk review –
  - Review of data and information provided in the VCS PD & MR and corresponding annexures.
  - Cross checking of the information with the sources without limitations to the information provided by the project proponents.
- Follow up actions –

  On-site inspection to physically inspect the project design including, but not limited to, monitoring mechanism. Information about the sampling plan, including sampling approach, important assumptions and justification of the chosen approach has been provided in section 2.4 Site inspection of this report.
- Interview with relevant personnel and farmers responsible for information given in the VCS PD & MR
- Reporting and closure of findings (CARs/CLs/FARs) and preparation of draft joint validation and verification report.

- Independent Technical Review of the draft report and final/revised documentation (e.g., VCS-Joint PD and MR, corresponding estimated ER calculations sheet and evidence).
- Issuance of the Final Validation and Verification Report to contracted PP (or authorized representatives).

No sampling was required to be used since all data was directly verified through evidence and supporting documents provided by PP.

## 2.2 Document Review

This joint validation & verification of the project is performed primarily as a document review of the VCS Joint PD & MR /1/ and associated documents as stated in detail in appendix I of this document. The assessment is performed by a validation & verification team using protocol. The cross checks between information provided in the VCS Joint PD & MR and information from sources other than those used, if available, the team's sectoral or local expertise and, if necessary, independent background investigations.

The joint validation and verification are performed primarily as a document review of the documents submitted at various stages of assessments. The review is performed by assessment team using dedicated protocols. The assessment team cross checks the information provided in the Joint Project Description & Monitoring Report/01/, documents and information from sources other than those used, if available, and also conducts independent background investigations. Earthood conducted a desk review as under –

- A review of the data and information presented to verify their completeness.
- A review of the monitoring plan, the monitoring methodology including applicable tool(s) and, where applicable, the applied standardized baseline, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures
- An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions

Furthermore, the validation team used additional documentation by third parties like host-party legislation, technical reports referring to the project design or to the basic conditions and technical data. The documents/evidence reviewed are included in Appendix III of this report.

## 2.3 Interviews

ESPL as a part of validation procedure conducted a comprehensive interaction with stakeholders. It was done during the site visit on 18/05/2023. It included interaction with project representative, project team of PP in order to assess the information included in the project documentation. The assessment team has interviewed the representatives of Project Proponent to confirm selected information and to clarify issues identified during document review.

The main topics covered during the interview are as follows:

- General aspects of the project
- Project Implementation
- Staff training procedures
- Calibration procedures
- Monitoring and measuring system
- Data collection, recording and archiving procedure
- QA/QC procedures
- VCS documentation
- Emission Reduction Calculations
- Input values, source documents of Additionality Assessment

The key personnel interviewed are summarized in the table given in Sec 2.4.

## 2.4 Site Visits

A physical site visit of the plant facility was undertaken by the assessment team on 18/05/2023. The Local expert was available on-site while the assessment team joined remotely. The farms were not visited due to an outbreak of disease in chickens and cattle farm; however, the farm records and the farm owner's declarations were reviewed on site/50/. The chicken and the cattle suffered from bird flu and foot and mouth disease respectively. The Site visit was conducted to assess the implementation and operation of the Project and to review evidence, and interview key personnel to confirm evidence associated with the project design, implementation, plant operations, environmental impacts, stakeholders etc/44/.

Based on the conducted interviews the validation team has thoroughly assessed the baseline scenario. The applicability of smallscale methodology AMS-III.D. was assessed through cross-checking the applicability conditions with the responses received from the farm owners. The validation team confirms that the responses received from the farm owners are in line with requirements stated in the methodology.

The key personnel interviewed are summarized in the table below:

| No | Interviewee | | | Date | Subject | Validation Team member |
|---|---|---|---|---|---|---|
| | Last name | First name | Affiliation | | | |
| 1. | Cakir | Bilgehan | Yılmazlar Madencilik | 18/05/2023 | • Description of the Project | Shreya Garg, Fikriye Seda |

<table>
<tr><td>2.</td><td>Dinc</td><td>Mehmet</td><td rowspan="2">Sanayi ve Ticaret Limited Şirketi<br><br>Project Owner</td><td rowspan="2"></td><td rowspan="2">• Eligibility Criteria, Applicability Conditions<br>• Baseline scenario<br>• Starting date and crediting period<br>• Additionality and investment analysis<br>• Emission Reductions calculations<br>• Compliance with relevant Laws<br>• Social & Environmental impacts</td><td rowspan="2">Atabek, Aayukta Singh</td></tr>
<tr><td>3.</td><td>Fidan</td><td>Ramazan</td></tr>
<tr><td>4.</td><td>Arikan</td><td>Dogukan</td><td rowspan="2">Life İklim ve Enerji Ltd. Şti.<br><br>Project Consultant</td><td rowspan="2">18/05/2023</td><td rowspan="2">• Project technology<br>• operation and maintenance,<br>• Project approval and implementation status,<br>• Agreements with the baseline farms<br>• Monitoring plan and monitoring arrangements<br>• Roles and responsibilities,<br>• Emergency procedures<br>• Social & Environmental impacts</td><td rowspan="2">Shreya Garg, Fikriye Seda Atabek, Aayukta Singh</td></tr>
<tr><td>5.</td><td>Bilir</td><td>Irem</td></tr>
<tr><td colspan="7">Local Stakeholders:</td></tr>
<tr><td>6.</td><td>Akdogan</td><td>Muzaffer</td><td>Mukhtar</td><td>18/05/2023</td><td>• Local Stakeholder Consultation<br>• Social and Environmental Impacts during construction and operation phase<br>• Establishment of the Grievance Mechanism</td><td>Shreya Garg, Fikriye Seda Atabek, Aayukta Singh</td></tr>
</table>

## 2.5 Resolution of Findings

The objective of this step is to identify, discuss and conclude on the issues related to the monitoring, implementation and operations of the registered project activity that could impair the capacity of the registered project activity to achieve emission reductions or influence the monitoring and reporting of emission reductions. This is done based on the desk review and interaction with Project Proponent and their site representative. The verification team prepares and/or updates a verification protocol (internal document) that records the conformities and non-conformities, which may be of following types.

CAR (Corrective Action Request) is raised if one of the following occurs:

- Non-compliance with the monitoring plan, the methodology or the standardized baseline are found in monitoring and reporting and has not been sufficiently documented by the project participants, or if the evidence provided to prove conformity is sufficient.
- Modifications to the implementation, operation and monitoring of the registered project activity has not been sufficiently documented by the project participants.
- Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions.
- Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the project participants.

Clarification request (CL) is raised if information is insufficient or not clear enough to determine whether the applicable CDM requirements have been met.

Forward action request (FAR) is raised during verification if the monitoring and reporting require attention and/or adjustment for the next verification period.

All CARs and CLs raised by the ESPL during verification shall be resolved prior to submitting a request for issuance. All the findings that are raised and communicated to project participant during the verification are included under Appendix 3. The section also includes the response, if provided, by the project proponent and an assessment by the verification team if it was closed out or otherwise.

During the validation and verification process, total 07 CARs and 04 CLs were raised and resolved satisfactorily. The list of CARs/CLs raised, and the response provided, the mean of validation, reasons for their closure and references to correction in the relevant documents are provided in Appendix IV of this report. There were no FARs raised for the next verification.

### 2.5.1 Forward Action Requests

No Forward Action Requests (FARs) were raised during this validation and verification of the project activity.

# 3 VALIDATION FINDINGS

## 3.1 Project Details

- **Project type, technologies and measures implemented, and eligibility of the project.**

  The Project is a biogas to energy project which aims to generate electricity annually from mixed waste. The Project consists of one engine with a capacity of 1,560 MWe. The plant has an installed capacity of 1.560 MWe.

In absence of the proposed Project, the animal manure was treated in uncovered anaerobic lagoons. The pProject, therefore, displaces an equivalent amount of electricity which would have otherwise been generated by fossil fuel dominant energy.

The plant collects both cattle manure generated at Kayi Farms and chicken litter from 11 other chicken farms on a daily basis through special sewage trucks equipped with close tanks to prevent any Odor and /or manure leakage and the same is fed into the anaerobic digester's facility. There is one solid waste receiving unit and one liquid receiving unit with the volume of 300 $m^2$ for each unit.

The start date of the Project is 29/09/2021 , which was the provisional acceptance protocol date of the gas-engine. This project adopts a renewable crediting period of 7 years with a maximum crediting period of 21 years i.e., twice renewable. The first crediting period is from 29/09/2021 to 28/09/2028, and the average yearly estimated emission reductions are 35,855 $tCO_2e$.

The Project consists of the following key equipment:

| Numbers of Units | Unit Name | VVB Assessment |
|---|---|---|
| 1 | Solid Waste Receiving Unit | VVB has assessed the following units with the help of single line diagram /45/ and it has been verified during the site visit/44/. |
| 2 | Liquid Waste Receiving Unit | |
| 3 | Material Mixing Unit | |
| 4 | Anaerobic Digesters | |
| 5 | Desulphurization and Gas Storage | |
| 6 | Separator Units | |
| 7 | Gas Engine and Flare Units | |

- **Project design, including eligibility criteria for grouped projects**

The Project is not a grouped project, hence eligibility criteria for grouped projects are not applicable.

- **Project proponent and other entities involved in the project –**

Yılmazlar Madencilik San. ve Tic. Ltd. Şti is the project proponents of the project which can be confirmed ownership documents /21b/nd the license provided /42/. No other entity is involved in this project.

- **Ownership –**

The Project is owned by Yılmazlar Madencilik San. ve Tic. Ltd. Şti. The ownership of the Project is verified through the following documents:

- Environmental Impact Assessment (EIA) Exemption Certificate issued by the Ministry of Environment and Urbanization in Türkiye/46/
- Electricity Generation License (EGL) issued by the Energy Market Regulatory Authority (EMRA) in Türkiye/42/

To avoid double counting, various registries like GS, GCC, CDM and they have been checked thoroughly and it has been confirmed that the project is not registered with any other GHG mechanism except VCS. PP has also provided the declaration/25/ regarding the same and it is found to be appropriate. The VVB has also reviewed the baseline surveys which confirms that, "no carbon credits will be claimed by the farm owners after the issuance process."

- **Project start date**

The start date of the Project is 29/09/2021 which is also the provisional acceptance protocol date of the first engine of the project. This was verified using provisional acceptance certificate/41/.

- **Project crediting period**

The crediting period of the Project is for 7 years (renewable twice for a total of 21 years). The first crediting period is from 29/09/2021 to 28/09/2028.

- **Project scale and estimated GHG emission reductions or removals**

The estimated annual emission reduction for the Project is 35,855 $tCO_2e$ which is less than 300,000 $tCO_2e$. Hence, the category is applicable under "Project"/40/.

- **Project location**

The "Yilmazlar Biogas Power Plant" is located in the Derbent Village of Tavşanlı District of Kütahya Province in Türkiye /39/.

Geo coordinates of the project location are:

| E | N |
|---|---|
| 29.353777222303226 | 39.628529407842706 |

The Geo-coordinates of farms that supply waste to the Project is given in the table below:

| Name of the Farm | E | N |
|---|---|---|
| Kayı Farm | 29°21'9.39" | 39°37'42.19" |
| As Tarım | 29°28'14.44" | 39°32'50.51" |
| Baş Tarım | 29°27'16.50" | 39°30'37.70" |
| Özkantar Gıda | 29°29'2.92" | 39°30'56.56" |
| Halil Yeşil Yumurta Üretim | 29°28'38.77" | 39°31'18.85" |
| Yoncalı Gıda | 29°26'14.36" | 39°31'23.64" |
| Yumurtalar Tarım | 29°28'42.65" | 39°31'19.31" |
| Emre Kayner | 29°28'35.76" | 39°31'16.43" |

| | | |
|---|---|---|
| Halil Küçük | 29°28'39.85" | 39°31'18.85" |
| Saydam Tarım | 29°28'35.04" | 39°31'21.16" |
| Bayraktar | 29°26'17.29" | 39°31'19.17" |
| Ali Aydın | 28°47'20.71" | 39°51'25.86" |

- **Conditions prior to project initiation**

The Project is set up to produce clean energy from renewable (biogas) energy; and to curb the emissions arising from animal manure decaying under anaerobic conditions in uncovered lagoons. The project is a greenfield project. There was no project installed prior to the commissioning of this project.

- **Project compliance with applicable laws, statutes and other regulatory frameworks**

"Yilmazlar Biogas Power Plant" is in compliance with current laws and regulations and there are no legal and/or regulatory requirements that prevent the Project's implementation. The project has obtained the EIA exemption certificate /46/ which was approved on 09/02/2021, and the Electricity Generation License /42/ which was issued on 20/05/2021.

- **Participation under other GHG programs:**

  - Projects registered (or seeking registration) under other GHG program(s)

Project is seeking registration under VCS only (Project ID – 4205). The PP has submitted the declaration/25/ which states that the net GHG emission reductions generated by the Project will not be used for compliance with any other emissions trading program or to meet binding limits on GHG emissions for the same monitoring period.

  - Rejection by other GHG programs

The project has not been rejected by other GHG programs.

- **Other forms of credit:**

  - **Emissions trading programs and other binding limits**

The project is not participating in other emission trading programs. Therefore, this condition is not applicable.

  - **Other forms of environmental credit sought or received and eligible to be sought or received**

The project seeking registration under the VCS (Project ID – 4205) and no other forms of environmental credits are sought or received as confirmed by the PP/21,25/.**Sustainable development contributions**The project uses non carbon technology and is helping to reduce the greenhouse gas emissions ($CH_4$ and $CO_2$) as compared to the business-as-usual scenario, created local employment (16 employees) and gave trainings /26/, and reduced other pollutants resulting from fossil fuel dependent power generation in host country.

| SDG Goal Number | SDG GOAL | Contributions |
|---|---|---|
| 7 | Affordable and Clean Energy | The Project is a biogas-to-energy project that generates thermal and electrical energy using renewable energy by capturing biogas mainly from animal manure via anaerobic digestion. By supplying renewable energy to the national grid of Türkiye, the Project contributes to increasing the share of renewable energy in the global energy mix and the proportion of the population with primary reliance on clean fuels and sustainable technology. The total amount of 81,900 MWh generated electrical energy will directly be fed to the Turkish national grid.<br><br>This was also verified by the VVB during the site visit/44/ and by reviewing other supportive documents (Such as, EIA Exemption Certificate/46/, Electricity generation Invoice/22/, and Electricity Generation License/42/). |
| 8 | Decent Work and Economic Growth | The Project will contribute to achieve higher levels of Türkiye's economic productivity. During the project's implementation, the project proponent employed 16 people. This was verified by the VVB during the site visit/44/, and by reviewing the social security records /26/. |
| 13 | Climate Action | The Project will reduce GHG emissions by capturing and utilizing methane. The Project sets to reduce an estimated average annual emission by 35,855 $tCO_2e$. This was verified by the ER sheets/40/ provided by the PD. |

**Additional information relevant to the project, including:**

- **Leakage management for AFOLU projects**

  Not applicable to the project activity.

- **Commercially sensitive information**

No commercially sensitive information has been excluded from the public version of the project description.

**Conclusion:**

In view of the assessment of VCS joint PD & MR/01/ and supporting documents as listed in Appendix III of this report, the assessment team is able to confirm that the description contained in the VCS joint PD & MR of the Project provides the reader with a clear understanding of the precise nature of the Project and the technical aspects of its implementation.

Consequently, ESPL confirms the project description of the project contained in the joint VCS PD & MR/01/ to be complete and accurate. The VCS joint PD & MR complies with the relevant forms and guidance for completing the VCS Joint PD & MR.

CL#01, CL#02, CAR#01, CAR#02 and CAR#03 were raised and resolved successfully.

## 3.2 Participation under Other GHG Programs

The Project is seeking registration under VCS programme only. The start date of the Project is 29/09/2021 which is also the provisional acceptance protocol date of the first engine of the project. Hence it can be confirmed that the project is complying with all the applicable VCS rules, including project start date and project crediting period requirements.

During the validation and verification process, the validation team reviewed the declaration/21/ submitted by the project proponent confirming that GHG emission reduction credits from the Project have not been registered under any other GHG program. It can thus be concluded that the Project is eligible to participate under the VCS Program.

## 3.3 Safeguards

### 3.3.1 No Net Harm

There is no negative impact to any socio-economic conditions of the region due to the project activity. No adverse environmental impact has been envisaged in the project activity, still all the necessary clearances from the state pollution control board, state forest department as well the ministry of environment and forests has been obtained.

The potential negative impacts are mitigated as emissions are monitored for compliance with air pollution regulations, and Odor control is ensured. Water and soil pollution are mitigated through recycling and impermeable surfaces. Solid and hazardous waste is managed according to regulations, and noise pollution is maintained below the permissible limit, preserving the well-being of nearby settlements.

It is worthy to note that the project activity has received EIA Exemption. The EIA exemption letter/46/ was referred by assessment team and it can be concluded that no adverse impact of the project has been anticipated.

This project activity will not involve any negative environmental or socio-economic impacts, as the project installed for generation of power using biomass which is a clean source of energy. Hence no mitigation measures are required.

### 3.3.2 Local Stakeholder Consultation

The local stakeholder consultation process has been described in detail, by the PP, in Section 2.2 of the VCS PD & MR/1/. The Project Participant correctly identified the relevant stakeholders such as the people residing in and around the Derbent Village of Tavşanlı District of Kütahya Province, local employees, representatives of the affected people residing in project area.

The stakeholder consultation process took place at Kayi Farms on November 15, 2022. Notices regarding the meeting were mailed to the Kutahya Provincial Directorate of Environment and Urbanization, Tavşanli Municipality, and environmental NGOs on Tuesday, November 2, 2022. Subsequently, telephone calls were made to village mukhtars for additional meeting details. The meeting included participation from local residents, the deputy mayor of Tavşanli Municipality, plant personnel, the project's advocate, and village mukhtars. A visual presentation was utilized to offer comprehensive information about the planned activity during the stakeholder consultation.

All the questions and concerns raised by stakeholders were answered satisfactorily by the PP and their representative and the same is documented under section 2.4 of the VCS PD & MR/1/. Stakeholders had no negative comments/complaints/grievances by the end of the meeting which could have any significant modification in the project description or its design. This information was cross-checked and confirmed through the interview with the Mukhtar who is the village headman and representative of the stakeholders who were not present at the public consultation meeting. VVB validation team was able to confirm that these stakeholders were part of the LSC meeting conducted by project developer and were found to be aware about the proceedings of the meeting. As per the VVB's assessment, PP has addressed all questions and responses of all the stakeholders with satisfaction.

For future grievances PP has established an ongoing grievance mechanism via Mukhtar. The grievance book is maintained at the plant site as verified during the onsite audit furthermore the contact details of the Yilmazlar biogas power plant representative have been provided to the Mukhtar office who is the representative for the village community and can place complaints or communicate grievances on behalf of the local community during the operation phase of the project.

CL#03 was raised and resolved successfully.

### 3.3.3 Environmental Impact

The EIA Exemption Certificate 46/ was approved and issued by the Ministry of Environment and Urbanization, Türkiye. The Project Identification document and Feasibility report /47,48/ prepared for the project covers all aspects of the project including Project Description and Characteristics, Current Environmental Characteristics of Project Location, Environmental Construction and Operation Phase of the Project Effects and Measures to be taken and Public Participation.

This Report has been evaluated by the relevant local government agencies and Ministry of Environment and Urbanization. After evaluation of the project and comments of the local agencies, the Ministry of Environment and Urbanization has concluded that project does not have significant environmental effects and the EIA assessment is positive for the project activities.

In accordance with the Turkish laws and regulations, EIA's approval shall be only made if a project subjected to the approval does not make any negative environmental and socio-economic impacts. Considering the fact that the proposed Project already obtained EIA approval by the Ministry of Environment and Urbanization in Türkiye, it is therefore possible to claim that there is no net harm linked to the project.

The validation team is of the opinion that the project complies with environmental regulations in Türkiye.

### 3.3.4 Public Comments

The PP hasn't received any public comments during the 30-day public comment period which commenced between 23/03/2023 to 23/04/2023.

### 3.3.5 AFOLU-Specific Safeguards

As the Project is a non-AFOLU project, this section is not applicable.

## 3.4 Application of Methodology

### 3.4.1 Title and Reference

The Project has applied CDM approved methodology AMS-III.D.- "Methane Recovery in animal manure management systems" (version 21.0). /7A/.

Title: GHG emission reductions through multi-site manure collection and treatment in a central plant Tools referenced in the applied methodology:

- CDM Tool 14 -" Project and leakage emissions from anaerobic digesters" (version 02.0)
- CDM TOOL 06 - "Project Emissions from Flaring" (version 04.0).
- CDM TOOL 05 - "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (version 03.0).
- CDM TOOL 03 - "Tool to calculate project or leakage CO2 emissions from fossil fuel combustion" (Version 03.0)
- CDM TOOL 08 – "Tool to determine the mass flow of a greenhouse gas in a gaseous stream." (Version 3.0)
- CDM TOOL 12 - "Project and leakage emissions from transportation of freight." (Version 01.1.0)

The Project also applies AMS.I.D.- “Grid-connected renewable electricity generation” (version 18.0). Sectoral scopes are listed below:

- CDM TOOL 03 - “Tool to calculate project or leakage CO2 emissions from fossil fuel combustion” (version 03.0)
- CDM TOOL 07 - "Tool to calculate the emission factor for an electricity system" (version 07.0)
- CDM TOOL 10 - “Tool to determine the remaining lifetime of equipment” (version 01.0)
- CDM TOOL 11 - “Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period.” (Version 03.0.1)

### 3.4.2 Applicability

The project “Yilmazlar Biogas Power Plant” complies with each applicability condition of the identified methodologies /7/. The same has been justified in the following table –

| AMS-III.D., "Methane Recovery in animal manure management systems "version 21.0 | | | |
|---|---|---|---|
| **No.** | **Applicability Conditions** | **Justification of applicability by PP** | **Means of assessment** |
| 1 | This methodology is only applicable under the following conditions:<br>(a) The livestock population in the farm is managed under confined conditions;<br>(b) Manure or the streams obtained after treatment are not discharged into natural water resources (e.g. river or estuaries), otherwise “AMS-III.H Methane recovery in wastewater treatment" shall be applied;<br>(c) The annual average temperature of baseline site where anaerobic manure treatment facility is located is higher than 5 °C;<br>(d) In the baseline scenario the retention time of manure waste in the anaerobic treatment system is greater than one | (a) The livestock population of the farms are under confined conditions. Moreover, these conditions are audited by the state since these conditions are mandatory by regulations.<br>(b) Manure or the streams obtained after treatment is not discharged into natural water resources.<br>(c) The annual average temperature of Kütahya, where the project is located, is 21.7 °C.<br>(d) Before the Project, anaerobic lagoons were used. The depth of the lagoons was more than 1 m. Moreover, the retention time was more than 1 month. | The livestock population is managed in the confined population, manure or the streams obtained are not discharged into natural water resources, the average temperature is 21.7 °C, the depth of the anaerobic lagoons used was more than 1m and the retention time was more than 1 month as was confirmed during the physical site visit /44/. This has also been confirmed from the baseline survey records. /50/<br><br>Therefore, the applicability criterion is met. |

| | | | |
|---|---|---|---|
| | month, and if anaerobic lagoons are used in the baseline, their depths are at least 1 m;<br>(e) No methane recovery and destruction by flaring or combustion for gainful use takes place in the baseline scenario. | (e) In the baseline, anaerobic lagoons were used. Therefore, no flaring and combustion for gainful use take place. | |
| 2 | The Project shall satisfy the following conditions:<br>(a) The residual waste from the animal manure management system shall be handled aerobically, otherwise the related emissions shall be taken into account as per relevant procedures of "AMS-III.AO Methane recovery through controlled anaerobic digestion". In the case of soil application, proper conditions and procedures (not resulting in methane emissions) must be ensured.<br>(b) Technical measures shall be used (including a flare for exigencies) to ensure that all biogas produced by the digester is used or flared.<br>( c ) The storage time of the manure after removal from the | (a) Residual waste from animal manure management system will be used as fertilizer. Since fertilizers are used aerobically, no methane emission will occur.<br>(b) Produced biogas will be burnt in order to generate electricity. In emergency situations, there is a flare to ensure all biogas is burnt. Therefore, the project complies with this condition.<br>(c) The cattle barn and BPP are in the same area, and chicken litter is brought from nearby farms. The manure will be taken into digesters just shortly after it is brought. Therefore, storage time will not exceed 45 days. | The residual waste from animal manure management system will be handled aerobically as fertilizers, further, the produced biogas will be burnt in order to generate electricity and also there is a flare to ensure all biogas is burnt.<br>Therefore, this applicability criterion is met. |

| | | | |
|---|---|---|---|
| | animal barns, including transportation, should not exceed 45 days before being fed into the anaerobic digester. If the project proponent can demonstrate that the dry matter content of the manure when remove | | |
| 3 | Projects that recover methane from landfills shall use “AMS-III.G Landfill methane recovery" and projects for wastewater treatment shall use AMS-III.H. Projects for composting of animal manure shall use “AMS-III.F Avoidance of methane emissions through composting". Project activities involving co-digestion of animal manure and other organic matters shall use the methodology “AMS-III.AO Methane recovery through controlled anaerobic digestion". | The project does not include methane from landfill, wastewater treatment, or composting of animal manure, or co-digestion of animal manure and other organic matters. Therefore, none of the mentioned methodologies is applicable. | The mentioned methodologies are not applicable because the project does not include methane from landfill, wastewater treatment, or composting of animal manure, or co-digestion of animal manure or other organic matter.<br><br>Therefore, this applicability criterion is not applicable. |
| 4 | Utilization of the recovered biogas in one of the options detailed in AMS-III.H is also eligible under this methodology. The respective procedures in AMS-III.H shall be followed in this regard. If the recovered biogas is used to power auxiliary equipment of the Project, it should be taken into account, accordingly, using zero as its emission factor; however, energy used for such purposes is not eligible as an SSC CDM Type I project component. | The electricity produced by the Project will be delivered to the grid. | The electricity produced during the Project will be delivered to the Turkish National Grid/43/.<br><br>Therefore, the applicability criterion is met. |

| | | | |
|---|---|---|---|
| 5 | New facilities (Greenfield projects) and project activities involving capacity additions compared to the baseline scenario are only eligible if they comply with the related and relevant requirements in the "General guidelines for SSC CDM methodologies". | The Project does not include any capacity additions. | The Project is a biogas power plant and does not include any capacity additions.<br><br>Therefore, this applicability condition is not applicable. |
| 6 | The requirements concerning demonstration of the remaining lifetime of the replaced equipment shall be met as described in the "General guidelines for SSC CDM methodologies". | The project is greenfield project, so there is no replaced equipment. | The project is a biogas power plant and does not include any replacement equipment.<br><br>Therefore, this applicability condition is not applicable. |
| 7 | Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO2 equivalent annually from all Type III components of the Project. | Expected emission reduction of the project is 35,855 $tCO_2e$/year which is less than the project. Therefore, the project complies with the condition. | The expected annual emission reductions are 35,855 $tCo_2e$/year which is less that 60kt CO2.<br><br>Therefore, this applicability condition is met. |
| **AMS-I.D., "Grid-connected renewable electricity generation "version 18.0** | | | |
| **No.** | **Applicability Conditions** | **Justification of applicability by PP** | **Means of assessment** |
| 1 | Install a Greenfield plant | The Project is a Greenfield project involving the setting up of a new power plant at a site where there were no power plants in the pre-project scenario. Therefore, the project implies the condition. | The project involves installation of a biogas power plant which is a greenfield project.<br><br>Therefore, the condition is met. |
| 2 | Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: | Proposed project is a biogas, so this condition is not applicable. | The proposed project is a biogas power plant project.<br><br>Therefore, this condition is not applicable. |

| | | | |
|---|---|---|---|
| 3 | If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM Project applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW. | The Project doesn't have any non-renewable components. The Project involves electricity generation by utilizing biogas produced from animal manure. Hence, the condition is not applicable. | The proposed project is a biogas power plant project.<br><br>Therefore, this condition is not applicable. |
| 4 | Combined heat and power (cogeneration) systems are not eligible under this category. | The Project is not a co-generation system, i.e.: the simultaneous generation of heat and electrical energy in one process. It consists in a biogas genset unit that produce electricity. The genset heat is recovered in a heat exchanger that transfer the heat to water that is utilized to maintain the required temperature conditions in the mesophilic reactors. | The proposed project is a biogas power plant project.<br><br>Therefore, this condition is not applicable. |
| 5 | In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the Project should be lower than 15 MW and should be physically distinct from the existing units. | The Project does not involve the capacity addition of renewable energy generation units at an existing renewable power generation facility since there was no renewable energy generation in the project site prior to Project. | The proposed project is a biogas power plant project and does not involve the capacity addition of renewable energy generation units.<br><br>Therefore, this condition is not applicable. |
| 6 | In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated, or replacement power plant/unit shall not exceed the limit of 15 MW. | The Project is not a retrofit, rehabilitation or replacement project. | The proposed project is a biogas power plant project and it's not a retrofit, rehabilitation or replacement project.<br><br>Therefore, this condition is not applicable. |

| | | | |
|---|---|---|---|
| 7 | In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with the procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration, other applicable Type-I methodologies such as "AMS-I.C.: Thermal energy production with or without electricity" shall be explored. | The recovered methane is used for electricity generation for supply to a grid. The baseline for the electricity component is in accordance with the procedure prescribed under this methodology. The recovered methane is not used for heat generation or cogeneration. | The recovered methane is used for electricity generation for supply to a grid. The recovered methane is not used for heat generation and cogeneration.<br><br>Therefore, this applicability condition is not applicable. |
| 8 | In case biomass is sourced from dedicated plantations, the applicability criteria in the tool "Project emissions from cultivation of biomass" shall apply. | The Project uses animal manure as a fuel which does not include any dedicated plantations. | The proposed project is a biogas power plant project which uses animal manure for the production of the same.<br><br>Therefore, this condition is not applicable. |

### 3.4.3 Project Boundary

As noted above, the baseline and monitoring methodology applied to the proposed Project are AMS-III.D. "Methane recovery in animal manure management systems", Version 21.0. and AMS-I.D. "Grid-connected renewable electricity generation" Version 18.0.

In accordance with these Methodologies, the project boundary includes:

- The livestock;
- Animal manure management systems (including centralized manure treatment plant);
- Facilities which recover and flare/combust or use methane;
- The project power plant and all power plants connected physically to the electricity system that the proposed project power plant is connected to.

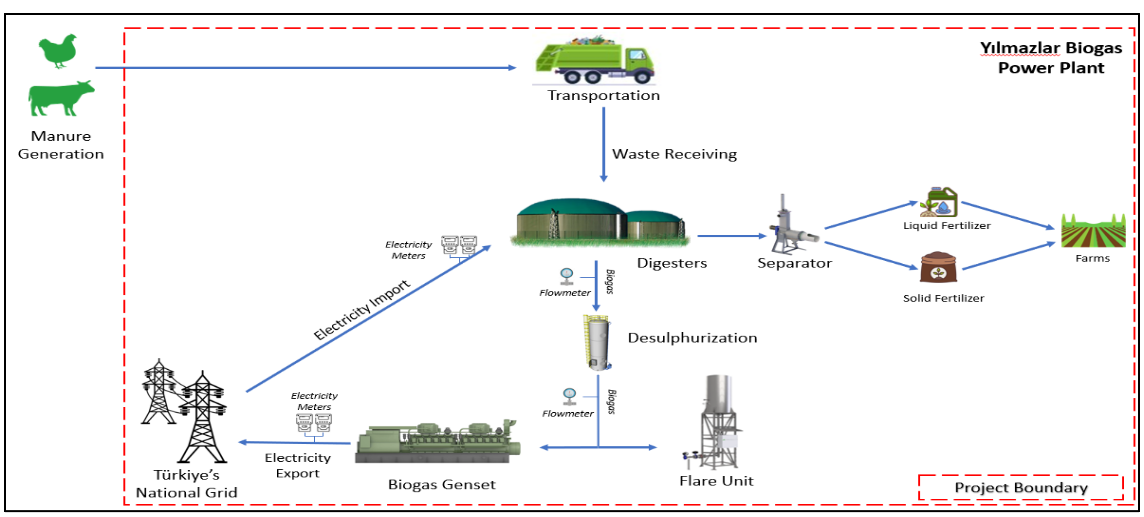
VVB could confirm this during the on-site audit/44/, from Google Earth, and also the commissioning certificate/18/. The assessment team was able to confirm that all the identified emission sources which are impacted by the Project are addressed by the approved methodology and can be seen in the Table below. The relevant GHG sources included in or excluded from the project boundary are shown on the Table below:

| Source | | Gas | Included? | Justification/Explanation |
|---|---|---|---|---|
| Baseline | Emissions from the manure treatment processes | $CO_2$ | No | CO2 emissions from the decomposition of organic waste are not accounted |
| | | $CH_4$ | Yes | The major source of emissions in the baseline |
| | | $N_2O$ | No | Excluded for simplification. This is conservative. |
| | Emissions from electricity consumption/ generation | $CO_2$ | Yes | Electricity is consumed from the grid in the baseline scenario |
| | | $CH_4$ | No | Excluded for simplification. This is conservative. |
| | | $N_2O$ | No | Excluded for simplification. This is conservative. |
| Project | Physical leakage of biogas | $CO_2$ | No | Excluded for simplification. This emission source is assumed to be very small. |
| | | $CH_4$ | Yes | The major source of emissions. Leakage from biogas digester has been considered. |
| | | $N_2O$ | No | Excluded for simplification. This emission source is assumed to be very small. |
| | Emissions from on-site electricity use | $CO_2$ | Yes | These emissions are accounted for since the Project consumes the electricity from the grid. |
| | | $CH_4$ | No | Excluded for simplification. This emission source is assumed to be very small. |
| | | $N_2O$ | No | Excluded for simplification. This emission source is assumed to be very small. |
| | Emissions from the flaring or | $CO_2$ | No | Flare system will be in use only emergency. If there is any flaring, emission reduction will be excluded during crediting period. |

| Source | | Gas | Included? | Justification/Explanation |
|---|---|---|---|---|
| | combustion of gas stream | $CH_4$ | No | Excluded for simplification. This emission source is assumed to be very small. |
| | | $N_2O$ | No | Excluded for simplification. This emission source is assumed to be very small. |
| | Emissions from incremental transportation distances | $CO_2$ | Yes | May be an important emission source. Hence, they are accounted |
| | | $CH_4$ | No | Excluded for simplification. This emission source is assumed to be very small. |
| | | $N_2O$ | No | Excluded for simplification. This emission source is assumed to be very small. |
| | Emissions from storage of manure | $CO_2$ | No | Excluded for simplification. This emission source is assumed to be very small. |
| | | $CH_4$ | Yes | Might be a major emission source. |
| | | $N_2O$ | No | Excluded for simplification. This emission source is assumed to be very small. |

The assessment team confirms that the Joint PDMR/1/ has correctly described the project boundary, including the physical delineation of the Project and complies with VCS standard version 4.4/04/.

The flow chart of the plant as depicted in the PDMR/1/ is given below:




## 3.4.4 Baseline Scenario

The VCS Joint PD & MR/1/ correctly follows the steps to determine the baseline scenario as per the applied methodologies AMS-III.D. and AMS-I.D. /7A, 7B/. The methodologies includes alternate scenarios for both, the owner of central treatment plant and the owner of livestock farms.

| Conditions | Justification provided by PP | VVB assessment |
|---|---|---|

<table>
<tr>
<td>Step 1 – Identify the various alternatives to the project proponent that deliver comparable levels of service, including the proposed Project undertaken without being registered as a CDM Project.</td>
<td>The Project is a greenfield project. According to the applied methodology for greenfield projects, the baseline scenario is selected from the complete set of the 2006 IPCC Guidelines for National Greenhouse Gas Inventories (Volume 4, Chapter 10, Table 10.14)[1].<br><br>Manure management systems are listed below:<br><br>Pasture, Range and Paddock<br><br>Solid Storage<br><br>Liquid/Slurry Storage<br><br>Anaerobic Lagoons<br><br>Pit Storage<br><br>Dry Lot<br><br>Daily Spread<br><br>Anaerobic Digestion – Biogas<br><br>Burned for Fuel</td>
<td>The alternate scenarios are identified from the 2006 IPCC Guidelines for National Greenhouse Gas Inventories (Volume 4, Chapter 10, Table 10.14), which are found to be valid and in-line with the applied methodology for greenfield projects.</td>
</tr>
<tr>
<td>Step 2 - List the alternatives identified in Step 1 that are in compliance with local regulations. If any of the identified baselines is not in compliance with local regulations, then exclude that alternative from further consideration).</td>
<td>Information for the alternatives is given in the table below.
<table>
<tr><th>IPCC Alternatives</th><th>Justification</th></tr>
<tr><td>Pasture, Range and Paddock</td><td>In accordance with “Regulation on Control of Odor Generating Emissions”, collected manure shall be stored before it is</td></tr>
</table>
</td>
<td>The assessment team has taken into consideration the following cases as alternatives to identify whether they are plausible or not:
<ul><li>Uncovered anaerobic lagoon is the current animal waste manure management system and is compliant with environmental and legal regulations in Türkiye and therefore</li></ul>
</td>
</tr>
</table>

[1] https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_10_Ch10_Livestock.pdf

| | | | | | |
|---|---|---|---|---|---|
| | | | spread to croplands. Therefore, the alternative is not applicable. | | applicable as a baseline and/or alternative scenario for the proposed Project.<br>• Liquid/Slurry is a common animal waste manure management for chicken farms. However, this kind of manure management system is not allowed due to the regulation that requires that manure shall be kept only on concrete made floor Pit storage under animal confinements is not allowed as per legal regulations of Türkiye.<br>• Solid storage as a manure management system is not in line with current regulations of Türkiye.<br>• Since animal farms in Türkiye are mandated to function in confined conditions, Dry lot as an alternative is not in compliance with regulatory requirements.<br>• In line with the legal regulations made on livestock farms, Daily spread is not allowed |
| | | Solid Storage | Solid storage system is not in line with current "Regulation on Livestock Farms" entered into force in 2006. | | |
| | | Liquid/Slurry Storage | This alternative does not comply with the local regulations since "Air Quality Regulation of Türkiye" states that liquid wastes shall be stored in impervious storage areas and transferred with closed containers. | | |
| | | Anaerobic Lagoons | In accordance with "Regulation on Livestock Farms" entered into force in 2006, animal | | |

| | | | | |
|---|---|---|---|---|
| | | | manure shall be kept in lagoons made in concrete so that it shall be leakage-proof. Therefore, the alternative complies with the local regulations. | due to the fact that untreated animal manure contains high nitrogen level and harms the organic quality of soil.<br>• Anaerobic digestion is the measure usen in the proposed Project. This alternative will be the Project not being registered with VERRA/VCS. Due to barriers such as technological barriers, lack of skilled labour to operate the facility and prevailing practice, as well as the investment barrier, this system is not applicable for the baseline scenario.<br>• Manure being burned for fuel is not in line with current regulations of Türkiye. |
| | | Pit Storage | Pit storage system was compliant with old regulations, but it is not allowed anymore. Animal farms with pit storage that was implemented before the new regulation, are asked to renew their storage system. | |
| | | Dry Lot | In accordance with “Regulation on Control of Odor Generating Emissions”, collected manure shall be stored | |

| | | | | |
|---|---|---|---|---|
| | | | before it is spread to croplands. Therefore, the alternative is not applicable. | |
| | | Daily Spread | In accordance with “Application Communique on Animal By-Products Regulation Not Used for Human Consumption”, collected manure shall be stored before it is spread to croplands. Therefore, the alternative is not applicable. | |
| | | Anaerobic Digestion - Biogas | Anaerobic digestion with biogas would associate with high investment costs and skilled manpower. There is also no legal requirement for aerobic treatment. | |
| | | Burned for Fuel | Burning animal | |

<table>
<tr>
<td></td>
<td><table><tr><td></td><td>manure is not allowed in accordance with local regulations.</td></tr></table></td>
<td></td>
</tr>
<tr>
<td>Step 3: Eliminate and rank the alternatives identified in Step 2 taking into account barrier tests specified in the "Guidelines on the demonstration of additionality of small-scale project activities".</td>
<td><table>
<tr><th>Alternative Scenario</th><th>Justification</th></tr>
<tr><td>The manure is disposed in an uncovered anaerobic lagoon;</td><td>This alternative is considered a realistic alternative since there is no regulation that restricts the scenario and applicable. The manure is disposed in an uncovered anaerobic lagoon. This is also the real baseline scenario.</td></tr>
<tr><td>Anaerobic Digestion - Biogas</td><td>This alternative is proposed Project. Although there are no regulatory restrictions to implement this scenario, the alternative is eliminated due to technological</td></tr>
</table></td>
<td>The VVB has assessed the identified alternatives that are found to be in compliance with the regulations. The two alternative scenarios identified are –<br>i. The manure being disposed in and uncovered anaerobic lagoon, this is the real case scenario, however, these lagoons lead to release of methane gas in the atmosphere which negates the role of alternative scenarios working positively towards the environment.<br>ii. The Project is implementation of the same, using anaerobic digesters and producing</td>
</tr>
</table>

| | | |
|---|---|---|
| | and investment barriers. | biogas via it. Therefore, this alternative scenario is found to be appropriate. |
| Step 4: If only one alternative remains that is not the proposed Project undertaken without being registered as a CDM Project and also corresponds to one of the baseline scenarios provided in the methodology, then the Project is eligible under the methodology. If more than one alternative remains that correspond to a baseline scenario provided in the methodology, choose the alternative with the lowest emissions as the baseline. | As a result of steps above, only one alternative scenario maintains all the requirements as baseline scenario. That is why, the proposed Project shall be eligible under AMS-III.D. methodology. | VVB considers this alternative as plausible because it represents the present situation. |

The VVB concludes that the VCS PD & MR /1/ conforms to the guidance given by VCS via VCS standard version 4.4/4/. The alternatives have been identified and justified in PD and the above section of this report.

The baseline scenario has been assessed by the validation team through physical site visit to the 12 farms ordinated by the PP and the supporting documents in order to confirm the description provided in the VCS PD/1/. The validation team confirms that the baseline scenario is in accordance with the applied small-scale methodologies AMS-III.D, v.21.0 and AMS-I.D, version 18.0.

### 3.4.5 Additionality

The project proponent has used para 15 and 16 for demonstration of additionality of the project. Following steps were used to check the additionality of the project -

In-line to para 15 of the approved methodology of AMS-III.D., version 21.0, "Project activities may demonstrate the additionality by showing that there is no regulation in the host country, applicable to the project site, that requires the collection and destruction of methane from livestock manure. If so, it is not required to apply the 'Guidelines on the demonstration of additionality of small-scale project activities.'

There are no legal regulations in Türkiye that requires collection and destruction of methane, however, there are related regulations present in the country –

| Relevant Law and Regulations for the Project | Comment | VVB Assessment |
|---|---|---|
| "Environmental Law[2]" dated 26/04/2006 numbered 2872 | The law prohibits the discharge, storage, transfer, and disposal of any kind of waste that does not meet the requirements according to Turkish regulations and standards. It includes the "polluter pays principle" and ensures that the producers take necessary measures to minimise waste.<br><br>The law doesn't mandate any alternative treatment processes for the animal manure management. | The VVB confirms that the project will take in necessary measures to reduce waste and also, wouldn't store, discharge, dispose and transfer any kind of waste that does not meet the requirements according to the "Environment Law".<br><br>Therefore, the project complies with the law. |
| Communiqué on Fermented Product Management by Mechanical Separation, Bio drying, Bio-methanization dated 10/10/2015 numbered 29498 | The purpose of this Communiqué is to ensure the management of biodegradable wastes without harming the environment and human health, to reduce the amount to be disposed of in landfills, to the technical criteria of mechanical separation, bio drying and bio-methanization facilities with material or energy recovery facilities, to the management of the fermented product obtained in bio-methanization facilities. | The project is a small-scale biogas power plant which will reduce the animal manure and cattle waste and will convert the same into biogas. It was found during the site visit that the waste are managed properly and about 24-26 tons of waste is bought in on daily basis in the plant and is managed in an anaerobic environment.<br><br>Therefore, the project complies with the law. |
| Regulation on Control of Odor Generating Emissions | The purpose of this Regulation is to regulate the administrative and technical procedures and principles for the control and reduction of odor-causing emissions. | The PP has covered the mixing unit with a tin cover to prevent odour which was confirmed during the site visit. |

[2] See: https://www.mevzuat.gov.tr/mevzuat?MevzuatNo=2872&MevzuatTur=1&MevzuatTertip=5

| | | |
|---|---|---|
| | | Therefore, the project is in compliance with this law. |
| Electricity Market Law[3] dated 20/02/2001 and numbered 4628 | The Law aims to ensure the development of a financially sound and transparent electricity market operating in a competitive environment under civil law provisions and the delivery of sufficient, good quality, low cost and environment-friendly electricity to consumers and to ensure the autonomous regulation and supervision of this market.<br><br>The regulation doesn't mandate any alternative treatment processes for electricity production. | The Project is the management of animal manure anaerobically and generating electricity which is further transmitted to the grid.<br><br>Therefore, the project is in compliance with the law. |
| Law on Utilisation of Renewable Energy Resources for the Purpose of Generating Electricity Energy[4] dated 10/05/2005 and numbered 5346 | The purpose of this Law is to expand the utilisation of renewable energy resources for generating electrical energy, to benefit from these resources in a secure, economical and qualified manner, to increase the diversification of energy resources, to reduce greenhouse gas emissions, to assess waste products, to protect the environment and to develop the related manufacturing sector for realising these objectives.<br><br>The Law brings an incentive of 13.3 $ cent/kWh for the electricity production from biomass. It also brings incentives for local equipment purchases such as turbines, engines, cogeneration systems etc.<br><br>The regulation does not mandate any alternative treatment processes for renewable electricity production. | The Project is the management of animal manure anaerobically and generating electricity which is further transmitted to the grid.<br><br>Therefore, the project is in compliance with the law. |

[3] See: https://www.mevzuat.gov.tr/MevzuatMetin/1.5.4628-20080709.pdf

[4] See: https://www.mevzuat.gov.tr/MevzuatMetin/1.5.5346.pdf

Since there are no legal regulations in Türkiye that requires collection and destruction of methane from livestock manure. Therefore, the project doesn't need to apply, "Guidelines on the demonstration of additionality of small-scale project activities" to demonstrate the additionality."

Further in-line to para 16 of AMS-III.D., version 21.0. clarifies that "This additionality condition also applies to Greenfield project activities. Furthermore, for project activities applying this methodology in combination with a Type I methodology, that has an energy component whose installed capacity is less than 5 MW, this procedure for additionality demonstration also applies to that component".

The installed capacity of the Biogas project plant is 1.560 MWe which is lower than 5MW. Therefore, in-line to para 15 and 16 of the applied methodology, AMS-III.D., the project is deemed additional.

### 3.4.6 Quantification of GHG Emission Reductions and Removals

BASELINE EMISSIONS:

**Baseline emissions in accordance to AMS.III.D, Version 21.0:**

Before the Project, the manure would decay anaerobically. Therefore, option (a) presented in AMS-III.D., version 21.0 paragraph 17 is utilised. Since there are both cattle and poultry farms that generates manure, emissions from both farms are considered separately and added into calculation.
In accordance with the option(b) baseline emissions are determined based on directly measured quantity of manure and its specific volatile solids content, as follows:

$$BE_y = GWP_{CH4} \times D_{CH4} \times UF_b \times \sum_{j,LT} MCF_j \times B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{Bl,j}$$

Equation 1

Where;

| | | |
|---|---|---|
| $BE_y$ | = | *Baseline emissions in year y (t $CO_2$e)* |
| $GWP_{CH4}$ | = | *Global Warming Potential (GWP) of $CH_4$ applicable to the crediting period (t $CO_2$e/t $CH_4$)* |
| $D_{CH4}$ | = | *CH4 density (0.00067 t/$m^3$ at room temperature (20 ºC) and 1 atm pressure)* |
| LT | = | *Index for all types of livestock* |
| j | = | *Index for animal manure management system* |
| $MCF_j$ | = | *Annual methane conversion factor (MCF) for the baseline animal manure management system j* |
| $B_{0,LT}$ | = | *Maximum methane producing potential of the volatile solid generated for animal type LT ($m^3$ $CH_4$/kg-dm)* |
| $N_{LT,y}$ | = | *Annual average number of animals of type LT in year y (numbers)* |

$VS_{LT,y}$ = *Volatile solids production/excretion per animal of livestock LT in year y (on a dry matter weight basis, kg-dm/animal/year)*

$MS\%_{Bl,j}$ = *Fraction of manure handled in baseline animal manure management system j*

$UF_b$ = *Model correction factor to account for model uncertainties (0.94)*

Yılmazlar biogas power plant receives waste from 12 different farms. List of the farms and information related to farms are given in the table below:

| Name of the Farm | Type of animal | Number of Animals | Average Weight of Animal (kg) | Rate of received manure |
|---|---|---|---|---|
| Kayı Farm | Dairy Cattle | 2,500 | 465 | 100% |
| As Tarım | Chicken Layer | 50,000 | 1.5 | 50% |
| Baş Tarım | Chicken Layer | 220,000 | 1.5 | 50% |
| Özkantar Gıda | Chicken Layer | 95,000 | 1.5 | 50% |
| Halil Yeşil Yumurta Üretim | Chicken Layer | 12,600 | 1.5 | 50% |
| Yoncalı Gıda | Chicken Layer | 350,000 | 1.5 | 50% |
| Yumurtalar Tarım | Chicken Layer | 28,000 | 1.5 | 50% |
| Emre Kayner | Chicken Layer | 18,500 | 1.5 | 50% |
| Halil Küçük | Chicken Layer | 15,000 | 1.5 | 50% |
| Saydam Tarım | Chicken Layer | 22,000 | 1.5 | 50% |
| Bayraktar | Chicken Layer | 330,000 | 1.5 | 50% |
| Ali Aydın | Chicken Layer | 70,000 | 1.5 | 100% |

To calculate $N_{LT,y}$, the following formula is used:

$$N_{LT,y} = N_{da,y} \times \frac{N_{p,y}}{365}$$

**Equation 2**

Where:

$N_{da,y}$ = Number of days animal is alive in the farm in the year y (numbers)

$N_{p,y}$ = Number of animals produced annually of type LT for the year y (numbers)

$N_{da,y}$ is considered as 365 values as per the plant logs which were obtained from the local farmers. Also, again as per the plant logs, $N_{p,y}$ values are taken as follows:

$N_{p,y}$ for dairy cattle = 2500 ; $N_{p,y}$ for chicken broiler = 1,211,100

Therefore, $N_{LT,y}$ values are 2500 for dairy cattle and 1,211,100 for chicken broiler.

To calculate $VS_{LT,y}$, the following formula is used:

$$VS_{LT,y} = \left(\frac{W_{site}}{W_{default}}\right) x VS_{default} x nd_y \qquad \textbf{Equation 3}$$

Where

$W_{site}$ = Average animal weight of a defined livestock population at the project site (kg)

$W_{default}$ = Default average animal weight of a defined population, this data is sourced from IPCC 2006 (kg)

$VS_{default}$ = Default value for the volatile solid excretion rate per day on a dry-matter basis for a defined livestock population (kg-dm/animal/day)

$nd_y$ = Number of days treatment plant was operational in year y

$W_{site}$ values are taken from the plant logs which were obtained from the local farmers. As per the logs, the values are 465 kg for dairy cattle and 1.5 kg for chicken broiler.

$W_{default}$ values are taken from IPCC Table 10A.5. The relevant values are 275 kg for dairy cattle and 0.7 kg for chicken broiler.

$VS_{default}$ values are taken from IPCC Table 10.13A. The relevant values are 10.7 kg-VS/(1000 kg animal mass)/day for dairy cattle and 17.7 kg-VS/(1000 kg animal mass)/day for chicken broiler.

$nd_y$ is taken as 365 days as per project proponent's declaration.

Provided below are calculated values of baseline emissions based on each farm:

$BE_{Kayı\ Farm,y}$=28 x 0,00067 x 0,94 x (0.73 x 0.24 x 2500 x 1816.0575 x 1)=14,027 $tCO_2e/y$

$BE_{As\ Tarım,y}$=28 x 0,00067 x 0,94 x (0.73 x 0.39 x 50000 x 7.7745 x 0.5)=976 $tCO_2e/y$

$BE_{Baş\ Tarım,y}$=28 x 0,00067 x 0,94 x (0.73 x 0.39 x 220000 x 7.7745 x 0.5)=4,294 $tCO_2e/y$

$BE_{Özkantar\ ,y}$=28 x 0,00067 x 0,94 x (0.73 x 0.39 x 95000 x 7.7745 x 0.5)=1,854 $tCO_2e/y$

$BE_{Halil\ Yeşil\ ,y}$=28 x 0,00067x 0,94 x (0.73 x 0.39 x 12600 x 7.7745 x 0.5)=246 $tCO_2e/y$

$BE_{Yoncalı\ ,y}$=28 x 0,00067 x 0,94 x (0.73 x 0.39 x 350000 x 7.7745 x 0.5)=6,831 $tCO_2e/y$

$BE_{Yumurtaları\ ,y}$=28 x 0,00067 x 0,94 x (0.73 x 0.39 x 28000 x 7.7745 x 0.5)=546 $tCO_2e/y$

$BE_{Emre\ Kayneri\ ,y}$=28 x 0,00067 x 0,94 x (0.73 x 0.39 x 18500 x 7.7745 x 0.5)=361 $tCO_2e/y$

$BE_{Halil\ Küçükı,y}$=28 x 0,00067 x 0,94 x (0.73 x 0.39 x 15000 x 7.7745 x 0.5)=293 $tCO_2e/y$

$BE_{Saydamı,y}$=28 x 0,00067 x 0,94 x (0.73 x 0.39 x 22000 x 7.7745 x 0.5)=429 $tCO_2e/y$

$BE_{Bayraktar,y}$=28 x 0,00067 x 0,94 x (0.73 x 0.39 x 330000 x 7.7745 x 0.5)=6,440 $tCO_2e/y$

$BE_{Ali\ Aydın,y}$=28 x 0,00067 x 0,94 x (0.73 x 0.39 x 70000 x 7.7745 x 1)=2,732 $tCO_2e/y$

As a result, BEy = 39,028 $tCO_2e/y$

**Baseline emission calculations according to AMS.I.D (version 18.0):**

In accordance with this methodology, baseline emissions include only $CO_2$ emissions from electricity generation in power plants that are displaced due to the Project. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,y}$$

Equation 4

Where;

| | | |
|---|---|---|
| $BE_y$ | = | *Baseline emissions in year y (t CO2)* |
| $EG_{PJ,y}$ | = | *Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM Project in year y (MWh)* |
| $EF_{grid,y}$ | = | *Combined margin CO2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t CO2/MWh)* |

The project is greenfield project so, in accordance with the paragraph 26 of AMS-I.D., the equation below shall be used for calculation of $EG_{PJ,y}$;

$$EG_{PJ,y} = EG_{PJ,facility,y}$$

Equation 5

| | | |
|---|---|---|
| $EG_{PJ,facility,y}$ | = | *Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh)* |

As per the applied methodology, the emission factor for the electricity generation $EF_{grid,CM,y}$ is determined according to "Tool to calculate the emission factor for an electricity system", version 07.0. According to the Tool, the "combined margin" emission factor is calculated through the following six steps as per the "Tool to calculate the emission factor for an electricity system", version 07.0:

Step (1): Identify the relevant electricity system;

Step (2): Choose whether to include off-grid power plant in the project electricity system (optional);

Step (3): Select a method to determine the operating margin (OM);

Step (4): Calculate the operating margin emission factor according to the selected method;

Step (5): Calculate the build margin (BM) emission factor;

Step (6): Calculate the combined margin (CM) emission factor

Step 1: Identify the relevant electricity system

According to the applicability of "Tool to calculate the emission factor for an electricity system", version 07.0, the relevant project electricity system should be identified by the project owners. The chosen electric power system is the Turkish National Grid, which is physically connected through transmission and distribution lines to the project activity. National grid is dominated by fossil fuels such as coal, diesel and natural gas.

Step 2: Choose whether to include off-grid power plants in the project electricity system

According to the applicable "Tool to calculate the emission factor for an electricity system", version 07.0, project owners may choose between the following two options to calculate the operating margin and build margin emission factor:

- Option I: Only grid power plants are included in the calculation
- Option II: Both grid power plants and off-grid power plants are included in the calculation

The project participants choose Option I and therefore only grid power plants are included in the calculation.

Step 3: Select a method to determine the operating margin (OM)

According to the applicable "Tool to calculate the emission factor for an electricity system", version 07.0, the operating margin emission factor ($EF_{grid,OM,y}$) is based on one of the following methods:

a) Simple OM; or

b) Simple adjusted OM; or

c) Dispatch data analysis OM; or

d) Average OM.

The simple OM method has been selected for calculation.

Step 4: Calculate the operating margin emission factor according to the selected method

The simple OM emission factor is calculated as the generation-weighted average $CO_2$ emissions per unit net electricity generation ($tCO_2/MWh$) of all generating power plants serving the system, not including low-cost/must-run power plants/units. The simple OM may be calculated by one of the following two options:

Option A: Based on the net electricity generation and a $CO_2$ emission factor of each power unit; or

Option B: Based on the total net electricity generation of all power plants serving and the fuel types and total fuel consumption of the project electricity system.

Ministry of Energy and Natural resources of Turkey has published the data for OM using simple OM calculation following the option A.

$EF_{grid,OM,y}$ = 0.7424 $tCO_2e/MWh$

For this approach (simple OM) to calculate the operating margin, the subscript m refers to the power plants/units delivering electricity to the grid, not including low-cost/must-run power plants/units. In addition, project owners has chosen to apply ex-post option for OM emission factor as described in the paragraph 42(b) of Tool 07, v07.0

<u>Step 5: Calculate the build margin (BM) emission factor</u>

In terms of vintage of data, project participants can choose between one of the following two options:

Option 1: for the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of PDD submission to the GCC Verifier for the project verification.

Option 2: for the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available.

For the proposed project activity, the project participants choose Option 1 in terms of availability of vintage of data. Based on 2020 data, Ministry of Energy and Natural resources of Turkey has calculated build margin and published the data:

$EF_{grid,OM,y}$ = 0.3680 $tCO_2e/MWh$

<u>Step 6: Calculate the combined margin emissions factor</u>

According to the applicable “Tool to calculate the emission factor for an electricity system”, version 07.0, the calculation of the combined margin (CM) emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM}$$

Equation 6

Where:

| | | |
|---|---|---|
| $EF_{grid,BM,y}$ | = | *Build margin $CO_2$ emission factor in year y ($t\ CO_2/MWh$)* |
| $EF_{grid,OM,y}$ | = | *Operating margin $CO_2$ emission factor in year y ($t\ CO_2/MWh$)* |
| $w_{OM}$ | = | *Weighting of operating margin emissions factor (per cent)* |

| | | |
|---|---|---|
| $w_{BM}$ | = | *Weighting of build margin emissions factor (per cent)* |

According to CDM Tool 07 "Tool to calculate emission factor for an electricity system" v07.0[5], paragraph 86(b) $w_{OM}$ and $w_{BM}$ for the first crediting period are given below:

$w_{OM} = 0.5$
$w_{BM} = 0.5$

As a result:

$EF_{grid,CM,y} = 0.7424 \times 0.5 + 0.3680 \times 0.5 = 0.5552$ $tCO_2/MWh$

So annual estimated emission reduction,

$BEy = EG_{pj,y} * EF_{grid,CM,y} = 11{,}700 \times 0.5552 = 6{,}495$ $tCO_2/year$

<u>PROJECT EMISSIONS:</u>

**Project Emissions as per AMS-III.D.**

The project emissions in year y are calculated for each alternative waste treatment process implemented in the Project as follows:

$$PE_y = PE_{PL,y} + PE_{flare,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y} \quad \text{Equation 7}$$

Where:

| | | |
|---|---|---|
| $PE_y$ | = | Project emissions in year y (t CO2e) |
| $PE_{PL,y}$ | = | Emissions due to physical leakage of biogas in year y (t $CO_2$e) |
| $PE_{flare,y}$ | = | Emissions from flaring or combustion of the biogas stream in the year y (t $CO_2$e) |
| $PE_{power,y}$ | = | Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (t $CO_2$e) |
| $PE_{transport,y}$ | = | Emissions from incremental transportation in the year y (t $CO_2$e) |
| $PE_{storage,y}$ | = | Emissions from the storage of manure (t $CO_2$e) |

[5] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf

**Project emissions from anaerobic digestion ($PE_{PL,y}$)**

$PE_{PL,y}$ is calculated according to “TOOL14: Project and leakage emissions from anaerobic digesters” as per AMS-III.D paragraph 21(b) stated.

As per paragraph 23 of the “TOOL14: Project and leakage emissions from anaerobic digesters”, The project emission associated with the anaerobic digester ($PE_{PL,y}$) equals to ($PE_{CH4,y}$) and is determined as follows:

$$PE_{CH4,y} = Q_{CH4,y} \times EF_{CH4,default} \times GWP_{CH4}$$

**Equation 8**

Where:

| | | |
|---|---|---|
| $PE_{CH4,y}$ | = | *Project emissions associated with the anaerobic digester in year y (t $CO_2$e)* |
| $Q_{CH4,y}$ | = | *Quantity of methane produced in the anaerobic digester in year y (t $CH_4$)* |
| $EF_{CH4,default}$ | = | *Project emissions from fossil fuel consumption associated with the anaerobic digester in year y (t $CO_2$e)* |
| $GWP_{CH4}$ | = | *Project emissions of methane from the anaerobic digester in year y (t $CO_2$e)* |

And $Q_{CH4,y}$ is calculated as follows:

$$Q_{CH4,y} = Q_{biogas,y} \times f_{CH4,default} \times \rho_{CH4}$$

**Equation 9**

| | | |
|---|---|---|
| $Q_{CH4,y}$ | = | *Quantity of methane produced in the digester in year y (t $CH_4$)* |
| $Q_{biogas,y}$ | = | *Amount of biogas collected at the digester outlet in year y ($Nm^3$ biogas)* |
| $f_{CH4,default}$ | = | *Default value for the fraction of methane in the biogas ($m^3$ $CH_4$ / $m^3$ biogas)* |
| $\rho_{CH4}$ | = | *Density of methane at normal conditions (t $CH_4$ / $Nm^3$ $CH_4$)* |

| Parameter | Value |
|---|---|
| $EF_{CH4,default}$ | 0.05 |
| $GWP_{CH4}$ | 28 |
| $f_{CH4,default}$ | 0.6 |
| $\rho_{CH4}$ | 0.00067 t/$m^3$ |

Project emissions from flaring are calculated as the sum of emissions from each minute m, based on the methane mass flow in the residual gas ($F_{CH4,RG,m}$) and the flare efficiency

($\eta_{flare,m}$), as follows:

$$PE_{flare,y} = GWP_{CH4} x \sum_{m=1}^{525600} F_{CH4,RG,m} \times \left(1-\eta_{flare,m}\right) \times 10^{-3}$$

**Equation 10**

Where:

$PE_{flare,y}$ = *Project emissions from flaring of the residual gas in year y ($tCO_2e$)*

$F_{CH4,RG,m}$ = *Mass flow of methane in the residual gas in the minute m (kg)*

$\eta_{flare,m}$ = *Flare efficiency in minute m*

| Parameter | Value |
|---|---|
| $\eta_{flare,m, biogas engine}$ | 100% |
| $\eta_{flare,m, flare unit}$ | 0% |

To calculate $F_{CH4,RG,m}$, Tool 08 (version 03.0) is used. Option 2: "Simplified calculation without measurement of the moisture content", Option A is chosen.

Flow measurement on a dry basis is not doable for a wet gaseous stream. Therefore, it is necessary to demonstrate that the gaseous stream is dry to use this option. There are two ways to do this:

(a) Measure the moisture content of the gaseous stream (CH2O,t,db,n) and demonstrate that this is less or equal to 0.05 kg H2O/m3 dry gas; or

(b) Demonstrate that the temperature of the gaseous stream (Tt) is less than 60ºC (333.15 K) at the flow measurement point.

As per (b), the temperature of the gaseous stream ($T_t$) is less than 60ºC (333.15 K) according to the thermocouple data.

$$F_{i,t} = V_{t,db} \times v_{j,t,db} \times \rho_{i,t}$$

with

$$\rho_{i,t} = \frac{P_n \times MM_i}{R_U \times T_n}$$

$F_{i,t}$ = Mass flow of greenhouse gas i in the gaseous stream in time interval t (kg gas/h)

$V_{t,db}$ = Volumetric flow of the gaseous stream in time interval t on a dry basis ($m^3$ dry gas/h)

$v_{i,t,db}$ = Volumetric fraction of greenhouse gas i in the gaseous stream in a time interval t on a dry basis ($m^3$ gas i/$m^3$ dry gas)

| | | |
|---|---|---|
| $\rho_{i,t}$ | = | Density of greenhouse gas i in the gaseous stream in time interval t (kg gas i/m³ gas i) |
| $P_n$ | = | Absolute pressure at normal conditions (Pa) |
| $MM_i$ | = | Molecular mass of greenhouse gas i (kg/kmol) |
| $R_u$ | = | Universal ideal gases constant (Pa.m3/kmol.K |
| $T_n$ | = | Molecular mass of the gaseous stream in a time interval t on a dry basis (kg dry gas/kmol dry gas) |

**Determination of project emissions from electricity consumption ($PE_{EC,y}$)**

According to Paragraph 16 of Tool 05, project emissions from the consumption of electricity are calculated based on the quantity of electricity consumed, an emission factor for electricity generation and a factor to account for transmission losses, as follows:

$$PE_{EC,y} = \sum_{j} EC_{PJ,j,y} \times EF_{EF,j,y} \times \left(1 + TDL_{j,y}\right)$$

**Equation 11**

Where:

| | | |
|---|---|---|
| $PE_{EC,y}$ | = | *Project emissions from electricity consumption in year y (t CO2 / yr)* |
| $EC_{PJ,j,y}$ | = | *Quantity of electricity consumed by the project electricity consumption source j in year y (MWh/yr)* |
| $EF_{EF,j,y}$ | = | *Emission factor for electricity generation for source j in year y (t $CO_2$/MWh)* |
| $TDL_{j,y}$ | = | *Average technical transmission and distribution losses for providing electricity to source j in year y* |
| j | = | *Sources of electricity consumption in the project* |

Therefore, $PE_{power}$ will be considered only if there is energy consumption without energy generation in monthly basis.

**Determination of the emission factor for electricity generation $EF_{EF,j,y}$**

According to the tool "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (version 03.0), for the project in the case of Scenario A: Electricity consumption from the grid, the project participants choose Option A1: Calculate the combined margin emission factor of the applicable electricity system, using the procedures in the latest approved version of the "Tool to calculate the emission factor for an electricity system" (version 07.0)

For the emission factor, the Turkish Ministry of Energy and Natural Resources publication, which indicates Türkiye's National Electric Grid Emission Factor for the year 2020, was used. The publication includes calculated Emission Factor values that are Operating Margin (OM), Growth Based Margin (Build

Margin-BM) and Combined Margin (CM) Emission Factors for the relevant year with the usage of the IPCC's Clean Development Methodology Tool 07-v07.0. For this calculation, information regarding the data set is given below in detail;

According to this publication;

- Operating Margin-OM; 0.7424 $tCO_2$/MWh
- Build Margin-BM; 0.3680 $tCO_2$/MWh
- Combined Margin-CM (other renewables than solar and wind); 0.5552 $tCO_2$/MWh[6]

**Determination of project emissions from incremental transportation in the year ($PE_{transport}$)**

In accordance with AMS-III.D., version 21.0., PEtransp,y is calculated as per the paragraph 14 of AMS-III.AO, version 01.0., which states project emissions due to incremental transport distances (PEtransp,y) are calculated based on the incremental distances between:

$$PE_{transp,y} = (Q_y / CT_y) * DAF_w * EF_{CO2,transport} + (Q_{res\text{-}waste,y} / CT_{res\text{-}waste,y}) * DAF_{res\text{-}waste} * EF_{CO2,transport} \quad \textbf{Equation 12}$$

Where:

| | | |
|---|---|---|
| $Q_{,y}$ | = | *Quantity of raw waste/manure treated and/or wastewater co-digested in the year y (tonnes)* |
| $CT_y$ | = | *Average truck capacity for transportation (tonnes/truck)* |
| $DAF_w$ | = | *Average incremental distance for raw solid waste/manure and/or wastewater transportation (km/truck)* |
| $EF_{CO2,transport}$ | = | *$CO_2$ emission factor from fuel use due to transportation (kgCO2/km, IPCC default values or local values may be used)* |
| $Q_{res,waste,y}$ | = | *Quantity of residual waste produced in year y (tonnes)* |
| $CT_{res\text{-}waste,y}$ | = | *Average truck capacity for residual waste transportation (tonnes/truck)* |
| $DAF_{res\text{-}waste}$ | = | *Average distance for residual waste transportation (km/truck)* |

The distances between farms and Yılmazlar BPP is given in the table below.

| Name of the Farm | Distance to the plant (km) |
|---|---|

6 https://enerji.gov.tr//Media/Dizin/EVCED/tr/%C3%87evreVe%C4%B0klim/%C4%B0klimDe%C4%9Fi%C5%9Fikli%C4%9Fi/EmisyonFaktorleri/BilgiFormu.pdf

| | |
|---|---|
| Kayı Farm | 0 |
| As Tarım | 18 |
| Baş Tarım | 25 |
| Özkantar Gıda | 25 |
| Halil Yeşil Yumurta Üretim | 25 |
| Yoncalı Gıda | 23 |
| Yumurtalar Tarım | 22 |
| Emre Kayner | 24 |
| Halil Küçük | 24 |
| Saydam Tarım | 24 |
| Bayraktar | 24 |
| Ali Aydın | 100 |

In accordance with studies poultry can defecate 3-4% of its weight in a day. So, for the calculation 4% has taken to be conservative and $Q_y$ has found accordingly.

**Project determine emissions from the storage of manure.**

Since the storage time of the manure after removal from the animal barns, including transportation, does not exceed 24 hours before being fed into the anaerobic digester in the proposed Project, $PE_{storage,y}$ equals to zero.

**Project Emissions as per AMS-I.D.**

As per the paragraph 39 of AMS-I.D., version 18.0., for most renewable energy project activities, PEy= 0. However, for the following categories of project activities, project emissions have to be considered following the procedure described in ACM0002 v21.0:

(a) Emissions related to the operation of geothermal power plants (e.g. non condensable gases, electricity/fossil fuel consumption);

(b) Emissions from water reservoirs of hydro power plants.

As per the paragraph 40 of AMS-I.D., CO2 emissions from on-site consumption of fossil fuels due to the Project shall be calculated using the latest version of the "Tool to calculate project or leakage $CO_2$ emissions from fossil fuel combustion". Also, the paragraph 41 of AMS-I.D. states that in case biomass is sourced from dedicated plantations, the procedures in the tool "Project emissions from cultivation of biomass" shall be used.

Since, none of the exception stated in the relevant paragraphs of AMS-I.D. mentioned above is applicable to the proposed Project, PEy as per AMS-I.D., version 18.0., shall be accounted as 0.

**Leakage:**

In accordance with AMS-III.D. the leakage emission of the project is calculated following the relevant procedure in the methodological tool "Project and leakage emissions from anaerobic digesters", version 02.0 and formula in the tool is given below:

$$LE_{AD,y} = LE_{storage,y} + LE_{comp,y}$$ **Equation 13**

Where:

| | | |
|---|---|---|
| $LE_{AD,y}$ | = | *Project emissions from electricity consumption in year y (t $CO_2$ / yr)* |
| $LE_{storage,y}$ | = | *Leakage emissions associated with storage of digestate in year y (t $CO_2e$)* |
| $LE_{comp,y}$ | = | *Leakage emissions associated with composting digestate in year y (t $CO_2e$)* |

Since solid digestate is not disposed in a SWDS or a stockpile, LEstorage,y are not considered for the solid digestate part. The solid digestate is distributed directly to the local farmers for free Therefore, $LE_{AD,y}$ equals to $LE_{storage,y}$ from liquid digestate. Leakage emissions for is calculated as per Tool 14 "Project and leakage emissions from anaerobic digesters" version 02.0. To calculate leakage emission from liquid digestate, Option 2 has chosen which presented in same tool, heading 6.2.1.3. According to the tool, leakage emission from liquid digestate is calculated by the following formula.

$$LE_{storage,y} = F_{ww,CH4,default} \times Q_{CH4,y} \times GWP_{CH4}$$ **Equation 14**

Where:

| | | |
|---|---|---|
| $LE_{storage,y}$ | = | *Leakage emissions associated with storage of digestate in year y (t $CO_2e$)* |
| $F_{ww,CH4,default}$ | = | *Default factor representing the remaining methane production capacity of liquid digestate (fraction)* |
| $Q_{CH4,y}$ | | *Quantity of methane produced in the digester in year y ($tCH_4$)* |
| $GWP_{CH4}$ | | *Global warming potential of CH4 ($tCO_2$ / $tCH_4$)* |

As per AMS-I.D. version 18.0, *"General guidance on leakage in biomass project activities shall be followed to quantify leakages pertaining to the use of biomass residues."*. Since this is not applicable for proposed project, leakage emission resulted from AMS-I.D. v.18 equals to zero.

<u>Net GHG Emission Reduction and Removals as per AMS-III.D</u>

To calculate the emission reductions as per AMS-III.D, following equation shall be applied as per paragraph 27 of the methodology:

$$ER_{y,ex\ post} = min[(BE_{y,ex\ post} - PE_{y,ex\ post}), (MD_y - PE_{power,y,ex\ post})]$$ **Equation 15**

Where:

| | | |
|---|---|---|
| $ER_{y,ex\ post}$ | = | *Emission reductions achieved by the Project based on monitored values for year y (t CO2e)* |

| | | |
|---|---|---|
| $BE_{y,ex\ post}$ | = | *Baseline emissions calculated using equation 1 (for projects using option in paragraph 17(a)) using ex post monitored values of $N_{LT,y}$ and if applicable $VS_{LT,y}$. For projects using option in paragraph 17(a), the ex post monitored values for $Q_{manure,j,LT,y}$ and $SVS_{j,LT,y}$ are used* |
| $PE_{y,ex\ post}$ | = | *Project emissions calculated using equation 6 using ex post monitored values of $N_{LT,y}$, $MS\%_{i,y}$, $MS\%_{l}$, All, $Q_{res\ waste,y}$ and if applicable $VS_{LT,y}$* |
| $MD_y$ | = | *Methane captured and destroyed or used gainfully by the Project in year y (t CO2e)* |
| $PE_{power,y,ex\ post}$ | = | *Emissions from the use of fossil fuel or electricity for the operation of the installed facilities based on monitored values in the year y (t CO2e)* |

Calculation of methane flared or combusted ($MD_y$) can only be calculated for verification and shall be calculated as follows:

$$MD_y = BG_{burnt,y} \times w_{CH4,y} \times D_{CH4} \times FE \times GWP_{CH4}$$

**Equation 16**

Where:

| | | |
|---|---|---|
| $BG_{burnt,y}$ | = | *Biogas flared or combusted in year y ($m^3$)* |
| $W_{CH4,y}$ | = | *Methane content in biogas in the year y (volume fraction)* |
| FE | = | *Flare efficiency in the year y (fraction)* |

If flared or combusted biogas data is not available, alternatively, biogas flared or combusted ($MD_y$) also can be calculated as follows:

$$MD_y = \frac{EG_y \times 3600}{NCV_{CH4} \times EE_y} \times D_{CH4} \times GWP_{CH4}$$

**Equation 17**

Where:

| | | |
|---|---|---|
| $EG_y$ | = | *Total electricity generated from the recovered biogas in year y (MWh)* |
| 3600 | = | *Conversion factor (1 MWh = 3600 MJ)* |
| $NCV_{CH4}$ | = | *NCV of methane (MJ/$Nm^3$) (use default value: 35.9 MJ/$Nm^3$)* |
| $EE_y$ | | *Energy conversion efficiency of the project equipment, which is determined by adopting one of the following criteria:* |

- *Specification provided by the equipment manufacture. The equipment shall be designed to utilize biogas as fuel, and efficiency specification is for this fuel. If the specification* *provides a range of efficiency values, the highest value of the range shall be used for the calculation.*

- *Default efficiency of 40 %.*

Summary of emission reduction as per AMS-III.D is given in the table below:

| Year | Estimated baseline emissions or removals ($tCO_2e$) | Estimated project emissions or removals ($tCO_2e$) | Estimated leakage emissions ($tCO_2e$) | Estimated net GHG emission reductions or removals ($tCO_2e$) |
|---|---|---|---|---|
| 29-September-2021 – 31-December-2021 | 9,944 | 662 | 1,802 | 7,480 |
| 2022 | 39,028 | 2,597 | 7,071 | 29,360 |
| 2023 | 39,028 | 2,597 | 7,071 | 29,360 |
| 2024 | 39,028 | 2,597 | 7,071 | 29,360 |
| 2025 | 39,028 | 2,597 | 7,071 | 29,360 |
| 2026 | 39,028 | 2,597 | 7,071 | 29,360 |
| 2027 | 39,028 | 2,597 | 7,071 | 29,360 |
| 01-January-2028 – 28-September-2028 | 29,084 | 1,935 | 5,269 | 21,880 |
| **Total** | **273,196** | **18,179** | **49,497** | **205,520** |
| **Average over the Crediting Period** | **39,028** | **2,597** | **7,071** | **29,360** |

<u>Net GHG Emission Reduction and Removals as per AMS-I.D</u>

In accordance with AMS-I.D., net emission reduction and removals has calculated by following formula:

$$ER_y = BE_y - PE_y - LE_y$$

Equation 18

$ER_y$ = *Emission reductions in year y ($tCO_2$)*

$BE_y$ = *Baseline Emissions in year y ($tCO_2$)*

$PE_y$ = *Project emissions in year y ($tCO_2$)*

$LE_y$ = *Leakage emissions in year y ($tCO_2$)*

As calculated in section 4.1 baseline emissions BE found as follows:

$BE_y = EG_{pj,y} * EF_{grid,CM,y} = 11,700 \times 0.5552 = 6,495\ tCO_2/year$

In accordance with paragraph 39 of AMS-I.D, project emission ($PE_y$) equals to zero and also leakage emission (LEy) was assumed as 0.

As a result

$ER_y = BE_y = 6,495\ tCO_2/year$

Summary of emission reduction as per AMS-I.D is given in the table below:

| Year | Estimated baseline emissions or removals ($tCO_2e$) | Estimated project emissions or removals ($tCO_2e$) | Estimated leakage emissions ($tCO_2e$) | Estimated net GHG emission reductions or removals ($tCO_2e$) |
|---|---|---|---|---|
| 29-September-2021 – 31-December-2021 | 1,672 | 0 | 0 | 1,672 |
| 2022 | 6,495 | 0 | 0 | 6,495 |
| 2023 | 6,495 | 0 | 0 | 6,495 |
| 2024 | 6,495 | 0 | 0 | 6,495 |
| 2025 | 6,495 | 0 | 0 | 6,495 |
| 2026 | 6,495 | 0 | 0 | 6,495 |
| 2027 | 6,495 | 0 | 0 | 6,495 |
| 01-January-2028 – 28-September-2028 | 4,823 | 0 | 0 | 4,823 |
| **Total** | **45,465** | **0** | **0** | **45,465** |
| **Average over the Crediting Period** | **6,495** | **0** | **0** | **6,495** |

**Summary of net emission reductions as per AMS-III.D and AMS-I.D are given int the table below.**

| Year | Estimated baseline emissions or removals ($tCO_2e$) | Estimated project emissions or removals ($tCO_2e$) | Estimated leakage emissions ($tCO_2e$) | Estimated net GHG emission reductions or removals ($tCO_2e$) |
|---|---|---|---|---|
| 29-September-2021 – 31-December-2021 | 11,598 | 662 | 1,802 | 9,134 |
| 2022 | 45,523 | 2,597 | 7,071 | 35,855 |
| 2023 | 45,523 | 2,597 | 7,071 | 35,855 |
| 2024 | 45,523 | 2,597 | 7,071 | 35,855 |
| 2025 | 45,523 | 2,597 | 7,071 | 35,855 |
| 2026 | 45,523 | 2,597 | 7,071 | 35,855 |
| 2027 | 45,523 | 2,597 | 7,071 | 35,855 |
| 01-January-2028 – 28 September 2028 | 33,923 | 1,936 | 5,269 | 26,718 |
| Total | 318,659 | 18,180 | 49,497 | 250,982 |
| Average over the Crediting Period | 45,523 | 2,597 | 7,071 | 35,855 |

The verification team confirms the following:

- All relevant assumptions and data are listed in the project description, including their references and sources.
- All data and parameter values used in the project description are considered reasonable in the context of the project.
- All estimates of the baseline emissions can be replicated using the data and parameter values provided in the project description.
- The validation team confirms that the methodology and the above referenced tools have been applied correctly to calculate baseline emissions, project emissions, leakage and net GHG emission reductions.

### 3.4.7 Methodology Deviations

The following methodology deviation has been sought by the PP:

As per the applied methodology AMS-III.D, v21.0, to estimate Wsite monitoring parameter, when IPCC values of VS are adjusted for site specific animal weight, sampling procedures shall be used. Since there is only one cattle farm with the animal number of 2500, all animals are weighted at the farm yearly and no sampling approach is needed for the dairy cattle farm. However, for the chicken broilers farms, the baseline surveys had to use to estimate Wsite parameter because owners of chicken farms do not allow the project owner to enter the farms since chickens are very susceptible to diseases. The baseline surveys were signed by the owners of the farms and these surveys have been provided to the VVB at the time of this joint validation/verification process.

### 3.4.8 Monitoring Plan

The Project has correctly applied the Approved Monitoring Methodology AMS-III.D, version 21.0 titled "Project emissions from electricity and fossil fuel consumption are determined by following the methodological tool "Project and leakage emissions from anaerobic digesters". The monitoring plan provides detailed information related to the collection and archiving of all relevant data needed to:

- Estimate or measure emissions occurring from GHG sources, sinks and reservoirs.
- Determine the baseline emissions.

The monitoring plan as per AMS-III.D, version 21.0 and AMS-I.D, version 18.0 has been clearly described in section 6.3 of the VCS PD & MR. It covers all the monitoring parameters required to monitor by the Project and emission reductions due to the Project accurately. The monitoring plan/procedure followed to measure the emission reduction is applied accurately and with a conservative approach.

**Parameters Determined ex-ante**

The following parameters are determined ex-ante and mentioned in section 5.1 of the PD:

| Parameter | Unit | Value applied | Means of assessment |
|---|---|---|---|
| $GWP_{CH4}$ | t $CO_2$ e/t $CH_4$ | 28 | This is a default value from IPCC fifth assessment report. Thus, the value is found acceptable. The value has been checked with the above-mentioned report /49/ is found to be consistently reported in the VCS PD&MR/01/. |
| $D_{CH4}$ | t $CO_2$ e/t $CH_4$ | 0.00067 | This is a default value from Methodological Tool 14 Project and leakage emissions from anaerobic digesters Version 02.0. Thus, the value is found acceptable. The value has been checked with the above-mentioned tool /8A/ and is found to be consistently reported in the VCS PD&MR/01/. |
| $UF_b$ | Dimensionless | 0.94 | This is a default value from CDM Methodology AMS-III.D -Methane recovery in animal manure |

| | | | |
|---|---|---|---|
| | | | management systems. Thus, the value is found acceptable. |
| $MCF_j$ | Fraction | 0.73 | This is a default value from IPCC fifth assessment report. Thus, the value is found acceptable. |
| $B_{0,LT}$ | $m^3CH_4/kg$-dm | $B_{0,LT,1}$ (Chicken-Layer): 0.39<br><br>$B_{0,LT,2}$ (Dairy Cattle): 0.24 | This is a default value from IPCC fifth assessment report. Thus, the value is found acceptable. |
| $VS_{LT,y}$ | Kg-dm/animal/year | Dairy Cattle – 1816.0575<br>Chicken Layer – 7.7745 | The default value for the volatile solid excretion per day on a dry-matter basis for a defined livestock population<br>This value is derived from the IPCC 2019 fifth assessment report. Thus, the value is found acceptable. |
| $EF_{CH4,default}$ | t CH4 leaked / t CH4 produced | 0.05 | This value is derived from IPCC 2019Report. Thus, the value is found acceptable. |
| $f_{CH4,default}$ | m3 CH4 / m3 biogas corrected to reference conditions | 0.6 | This value is derived from Tool 14 Project and leakage emissions from anaerobic digesters Version 02.0 and the value was derived based on reported values from registered projects and research papers (Davidsson, 2007). Thus, the value is found acceptable. |
| $EF_{grid,CM,y}$ | tCO2/MWh | 0.5552 | This value is derived from the Türkiye emission factor factsheet, 2020.<br>Thus, the value is found acceptable |
| $EF_{CO2,transport}$ | kgCO2/km | 6.37 | Default value derived from Project and leakage emissions from road transportation of freight v.1.0. Thus, the value is found acceptable |
| $TDL_{j,y}$ | -- | 3% | Default value derived from Methodological tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (version 03.0). Thus, the value is found acceptable |

| | | | |
|---|---|---|---|
| $NCV_{CH4}$ | MJ/Nm³ | 35.9 | The value is a default value derived from AMS-III.G. "Landfill methane recovery" (version 10.0). Thus, the value is found acceptable |
| $EE_y$ | % | 40% | The value is a default value derived from AMS-III.G. "Landfill methane recovery" (version 10.0). Thus, the value is found acceptable. |
| $F_{ww,CH4,default}$ | Fraction | 0.20 | The value is a default value derived from Tool14 "Project and leakage emissions from anaerobic digesters" (version 02.0).<br>Thus, the value is found acceptable. |
| $W_{default}$ | kg | 275 kg for dairy cattle<br><br>0.7 kg for chicken broiler | This value is derived from the IPCC 2019 fifth assessment report. Thus, the value is found acceptable. |
| $VS_{default}$ | kg-VS/(1000 kg animal mass)/day | 10.7 kg-VS/(1000 kg animal mass)/day for dairy cattle<br><br>17.7 kg-VS/(1000 kg animal mass)/day for chicken broiler | This value is derived from the IPCC 2019 fifth assessment report. Thus, the value is found acceptable. |
| $P_n$ | Pa | 101,325 | The value is derived using Tool 08, "Tool to determine the mass flow of a greenhouse gas in a gaseous stream." (Version 3.0). Thus, the value is found acceptable |
| $T_n$ | K | 273.15 | The value is derived using Tool 08, "Tool to determine the mass flow of a greenhouse gas in a gaseous stream." (Version 3.0). Thus, the value is found acceptable |
| $MM_i$ | kg/kmol | 16.04 | The value is derived using Tool 08, "Tool to determine the mass flow of a greenhouse gas in a gaseous stream." (Version 3.0). Thus, the value is found acceptable |
| $R_u$ | Pa.m3/kmol.K | 0,008314 | The value is derived using Tool 08, "Tool to determine the mass |

| | | | |
|---|---|---|---|
| | | | flow of a greenhouse gas in a gaseous stream.” (Version 3.0). Thus, the value is found acceptable |

The parameters for ex-ante are found to be aligned with the applied methodology/8/ and are found to correct, justified and consistent through all documents.

**Data/parameters monitored:**

| S.No. | Parameter | Value | Monitoring frequency | Means of assessment |
|---|---|---|---|---|
| 1. | $EG_{PJ,y}$<br>Quantity of net electricity generation supplied by the project plant/unit to the grid in year y<br>MWh/year | 11,700 MWh<br>(Ex-ante estimate) | Monthly | The parameter will be monitored continuously by the energy meters installed at the plant site/35/ and to be cross verified from the EPIAs records |
| 2. | $EG_y$<br>Quantity of gross electricity generated by the project plant<br>MWh/year | Will be monitored Ex-post | Continuous measurement, hourly recording, monthly reported. | The parameter will be monitored continuously by using the SCADA system on the biogas genset |
| 3. | $N_{da,y}$<br>Number of days animal is alive in the farm in the year y | 365 | Monthly | This parameter will be monitored Monthly and will be verified from the Plant’s log/35/. |
| 4. | $N_{p,y}$<br>Number of animals produced annually of type LT for the year y | 2500 - Dairy cattle,<br>1,211,100 - Chicken layers | Monthly | The values have been derived as per the agreements signed with the farmers and the number of cattle and chickens present in the farm. |
| 5. | $MS\%_{Bl,j}$<br>Fraction of manure handled | 100% - Kayı Farm and Ali Aydın farm | Annually | The values have been derived on the basis of the agreements signed with the farmers and also can be |

| | | | | |
|---|---|---|---|---|
| | in system j in the Project | 50% - for rest of the farms | | cross-verified with the plant logs/35/. |
| 6. | $W_{site}$<br><br>Average animal weight of a defined livestock population at the project site | 465 kg for diary cattle<br><br>1.5 kg for chicken layer | Continuously | The parameter will be monitored regularly and has been derived from the plant logs /35/. |
| 7. | $Q_{biogas}$<br><br>Amount of biogas collected at the digester outlet in year y<br><br>Nm3 biogas | 3,704,429 Nm3 | Annually | The parameter will be monitored continuously and will be recorded monthly and yearly and will be recorded in the SCADA system daily. |
| 8. | $W_{CH4}$<br><br>Methane content in biogas in the year y<br><br>% | 60% | Continuous measurement, monthly readings | The parameter will be monitored continuously as flow meter at a 90/10 confidence/precision level by following the "Standard for sampling and surveys for CDM project activities and Programme of Activities". |
| 9. | FE<br><br>Flare Efficiency<br><br>% | 0% for flare unit<br><br>100% for biogas engine | - | The value for this parameter is considered zero considering the conservative approach. |
| 10. | $F_{CH4,RG,y}$<br><br>Mass of methane sent to flare in year y<br><br>$tCH_4$ | 0 | Continuous | The parameter will be monitored continuously and will be monitored via a flow meter. |
| 11. | $Q_y$ and $Q_{res\ waste,y}$<br><br>Quantity manure brought to the plant<br><br>Tons | - | Monthly | This will be monitored monthly on the basis of on-site data records available at the site. |
| 12. | $CT_y$ and $CT_{res\ waste,y}$ | 26tons/ average | Monthly | This parameter will be monitored monthly on the |

| | | | | |
|---|---|---|---|---|
| | Average truck capacity for transportation<br><br>Tons/truck | | | basis of om-site records available at the plant. |
| 13. | $DAF_w$ and $DAF_{res\ waste}$<br><br>Average incremental distance for raw solid or product transportation<br><br>Km/truck | - | Annually | This parameter will be measured annually via Map applications available in the region. |
| 14. | $nd_y$<br><br>Number of days that the animal manure management system was operational<br><br>days | 365 | - | This parameter is used to calculated the baseline emissions. |
| 15. | $V_{t,db}$<br><br>Volumetric flow of the gaseous stream in time interval t on a dry basis<br><br>$m^3$ dry gas/h | - | Continuous | This parameter will be measured continuously by using flow meter and will be used to calculate the project emissions. |
| 16. | $v_{i,t,db}$<br><br>Volumetric fraction of greenhouse gas i in a time interval t on a dry basis<br><br>$m^3$ gas i/$m^3$ dry gas | - | Continuous | This parameter will be measured continuously by using gas analyzer and will be used to calculate the project emissions. |

Detailed responsibilities and authorities for project management, monitoring procedures, calibration procedures and QA/QC procedures have been presented and were verified during follow up interviews. The detailed monitoring practice is considered appropriate and the implementation of these will enable subsequent verification of the project's emission reductions.

All relevant data will be archived electronically and further maintained for the entire crediting period plus two years. Based on the above assessment the validation team concludes that the PP is capable of implementing the monitoring plan and hence confirms compliance of VCS guidelines/3/ and the applied methodology/7A, 7B/.

## 3.5 Non-Permanence Risk Analysis

This is not applicable to the Project as the Project is not an AFOLU project.

# 4 VERIFICATION FINDINGS

## 4.1 Accuracy of GHG Emission Reduction and Removal Calculations

Data and parameters available at the time of validation are assessed and presented in section 3.4.8. of this report. Data/parameter monitored are also assessed and presented in section 3.4.8. of this report.

In conclusion, VVB reviewed all the evidence submitted by PP related to GHG emission reduction calculations and confirmed that all the parameters are correctly applied. Default values used in the calculation were identified correctly. The emission reduction calculation has been done in line with the applied methodologies /7A,7B/. The assessment of sampling approach followed by PP was also checked and was found to be in line with the requirements of applied methodology. The assessment team also found that the monitoring plan is sufficiently provided in the PD&MR and the same was confirmed during site visit.

## 4.2 Quality of Evidence to Determine GHG Emission Reductions and Removals

The assessment team confirms that the calculations and data are authentic. The assessment team has reviewed the EPIAS Records, flow meter records, flare records and the ER calculation sheet submitted for validation and verification and confirms them to be appropriate /22//33//40//51/. The assessment team has checked the quality and maintenance of the supporting documents during the on-site visit to confirm the authenticity of the documents and to check the appropriate calculations. The assessment team confirms that the proper evidence is available for the whole monitoring period and the same is verifiable and the data collection system meets the requirement of the monitoring plan and the applied methodology according to the assessment carried out.

The assessment team confirms the quality of evidence to determine the GHG reductions are satisfactory and the detailed information regarding the roles and responsibilities have been provided in the Joint PD & MR.

# 5 VALIDATION AND VERIFICATION OPINION

Earthood Services Private Limited (ESPL) has been contracted by Arvind Ltd. (PP) to validate and verify the Project “Yılmazlar Biogas Power Plant Project” (project ID 4205). The validation and verification assessment has been completed and concluded after thorough assessment of PD & MR, ER calculation sheet, evidence in line with the requirements of VCS Standard, v4.4 and applied methodology AMS- III.D, version 21.0.

In the course of the validation 04 Clarification requests (CLs) and 07 Corrective Action Requests (CARs were raised and successfully closed and no FVR. The review of the revised PD & MR versions, the supporting documents and subsequent follow-up discussions provided adequate evidence to conclude the closure of specified CARs and CLs.

No limitations or doubts were identified related to the validation of the project. The conservative approach and methodological choices used in the project design make it very likely that the project will meet the projected emissions reduction target.

Earthood Services Private Limited has reviewed the project description documents and subsequently carried out site visit interviews to confirm the fulfilment of stated criteria. The Project has correctly applied the baseline and monitoring methodology AMS.III.D.- “Methane Recovery in animal manure management systems” (version 21.0) which is an approved methodology under CDM and is acceptable under VCS Version 4.4. The baseline has been determined in accordance with the stated approved baseline methodology.

As summary the validation and verification team able to conclude that:

- The project is in line with all relevant host country criteria (Türkiye) and all relevant VCS version 4.4 program guidelines requirements.
- The project additionality is sufficiently justified in the VCS PD & MR.
- The monitoring plan is transparent and adequate and in line with applied baseline and monitoring methodology of AMS-III.D, version 21.0 and AMS-I.D, version 18.0.
- The calculation formulae and use of parameter for the project emission reductions estimation are transparent and in line with the requirement of the applied methodology. The ex-ante projection of emission reductions given is found to be appropriate, conservative and in line with the requirement. The estimated Emission Reductions during the crediting period by the Project is expected to be 250,983 $tCO_2e$ over the 7-year project crediting period.

The conclusions of this report show that the project, as it was described in the project documentation and monitoring report, is in line with all criteria applicable for the validation and verification as outlined under VCS Standard v4.4.

ESPL here concluded that the PD & MR and its estimated emissions reduction calculations are in with all criteria applicable for the validation and verification against the VCS standard v4.4. The Project is found to be eligible under Sectoral Scope 1, “Renewable Energy” and Sectoral Scope 13 “Waste Handling and Disposal”. The reasonable level of assurance has been attained in this validation and verification assessment. PP has provided sufficient evidence and applied valid version of VCS documents. It is concluded that the project is likely to achieve estimated GHG emission reduction or removals. Therefore, ESPL is able to certify that the project meets all relevant requirements of the above-defined criteria and recommend registration of the Project and issuance of carbon credits.

The net estimated ERs are mentioned below:

| Year | Estimated baseline emissions or removals ($tCO_2e$) | Estimated project emissions or removals ($tCO_2e$) | Estimated leakage emissions ($tCO_2e$) | Estimated net GHG emission reductions or removals ($tCO_2e$) |
|---|---|---|---|---|
| 29-September-2021 – 31-December-2021 | 11,598 | 662 | 1,802 | 9,134 |
| 2022 | 45,523 | 2,597 | 7,071 | 35,855 |
| 2023 | 45,523 | 2,597 | 7,071 | 35,855 |
| 2024 | 45,523 | 2,597 | 7,071 | 35,855 |
| 2025 | 45,523 | 2,597 | 7,071 | 35,855 |
| 2026 | 45,523 | 2,597 | 7,071 | 35,855 |
| 2027 | 45,523 | 2,597 | 7,071 | 35,855 |
| 01-January-2028 – 28 September 2028 | 33,923 | 1,936 | 5,269 | 26,718 |
| Total | 318,659 | 18,180 | 49,497 | 250,982 |
| Average over the Crediting Period | 45,523 | 2,597 | 7,071 | 35,855 |

The achieved ERs are mentioned below:

| Year | Baseline emissions or removals ($tCO_2e$) | Project emissions or removals ($tCO_2e$) | Leakage emissions ($tCO_2e$) | Net GHG emission reductions or removals ($tCO_2e$) |
|---|---|---|---|---|
| 29-September-2021 – 31-December 2021 | 2,807 | 0 | 506 | 2,301 |
| 01-January-2022 – 31-December 2022 | 33,941 | 0 | 6,087 | 27,854 |
| 01-January-2023 – 31 March 2023 | 9,930 | 933 | 2,002 | 6,995 |
| Total | **46,678** | **933** | **8,595** | **37,150** |

Comparison of the ERs is provided below:

| Ex-ante emissions reductions /removals | Achieved emissions reductions/ removals | Percent difference | Justification for the difference | VVB assessment |
|---|---|---|---|---|
| **53,929 $tCO_2e$.** | 37,150 $tCO_2e$ | -45.2% | Ex-ante calculations are much higher than ex-post emission reductions. The reason behind is that the project could not be operational and generate electricity for a while although the plant is commissioned. Moreover, the plant is slowly increasing its generated electricity as well as methane destruction. It is expected that the percent difference will be lower in following monitoring periods since the Project | This project is in its initial stage of implementation, and the plant is gradually increasing its electricity generation as well as methane destruction. The VVB confirms that the plant is operational and is gradually increasing its efficiency. |

| | | | | |
|---|---|---|---|---|
| | | | approximately reached its full capacity. | |

Approved by:

Kaviraj Singh

Managing Director

Earthood Services Private Limited

Date: 01/02/2023

Place: Gurgaon, Haryana

# APPENDIX I: ABBREVIATIONS

| Abbreviations | Full texts |
|---|---|
| BE | Baseline Emission |
| BM | Build Margin |
| CAR | Corrective Action Request |
| CDM | Clean Development Mechanism |
| CL | Clarification Request |
| CM | Combined Margin |
| $CO_2$ | Carbon di oxide |
| CP | Crediting Period |
| DR | Desk Review |
| EB | Executive Board |
| EF | Emission Factor |
| ER | External Resource |
| EIA | Environmental Impact Assessment |
| ESPL | Earthood Services Private Limited |
| FAR | Forward Action Request |
| GHG | Green House Gas |

| | |
|---|---|
| MWh | Mega Watt Hour |
| MR | Monitoring Report |
| PD | Project Description |
| PP | Project Proponent |
| $tCO_{2e}$ | Tonnes of Carbon di oxide equivalent |
| TA | Technical Area |
| UNFCCC | United Nations Framework Convention on Climate Change |
| VCS | Verified Carbon Standard |
| VCU | Verified Carbon Units |
| VVB | Validation and Verification Body |

# APPENDIX II: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWER

| **Competence Statement** | | | |
|---|---|---|---|
| **Name** | Shreya Garg | | |
| **Country** | India | | |
| **Education** | M.Sc. (Climate Science & Policy), TERI University | | |
| **Experience** | 9 Years + | | |
| **Field** | Climate Change | | |
| **Approved Roles** | | | |
| **Team Leader** | YES | | |
| **Validator** | YES | | |
| **Verifier** | YES | | |
| **Methodology Expert** | AMS.I.A., AMS.I.C., AMS.I.D., AMS.I.F., AMS.II.D., AMS.II.G., AMS.II.J., AMS.III.AV., AMS.III.BL, ACM0002, ACM0012 | | |
| **Local expert** | YES (India) | | |
| **Financial Expert** | NO | | |
| **Technical Reviewer** | YES | | |
| **TA Expert** | YES (TA 1.2, TA 3.1) | | |
| | | | |
| **Reviewed by** | Shifali Guleria | **Date** | 26/04/2022 |
| **Approved by** | Deepika Mahala | **Date** | 26/04/2022 |

| **Competence Statement** |
|---|

| | | | |
|---|---|---|---|
| **Name** | Kaviraj Singh | | |
| **Education** | Ph.D. (Environmental Engineering), IIT Delhi<br>Masters (Energy & Environmental), DAVV Indore | | |
| **Experience** | 15 Years + | | |
| **Field** | Climate Change & Environment | | |
| **Approved Roles** | | | |
| **Team Leader** | YES | | |
| **Validator** | YES | | |
| **Verifier** | YES | | |
| **Methodology Expert** | AMS-I.D., AMS-II.D., ACM0006, AMS-I.A., AMS-I.C., AMS-II.B., AMS-III.H, ACM0002, ACM0001, AM0080, ACM0018, AM0056, AM0073<br>VM0042, AMS-III.G, AMS-III.AF., VM0032, VM0018, ACM0010, ACM0022, AMS-III.D, AMS-III.F and AMS-III.A.Q | | |
| **Local expert** | YES (India) | | |
| **Financial Expert** | YES | | |
| **Technical Reviewer** | YES | | |
| **TA Expert (X.X)** | YES (TA 1.1, TA 1.2, TA 3.1, TA 13.1, TA 13.2) | | |
| | | | |
| **Reviewed by** | Shifali Guleria (Quality Manager) | **Date** | 02/02/2023 |
| **Approved by** | Deepika Mahala (Technical Manager) | **Date** | 02/02/2023 |

| **Competence Statement** | |
|---|---|
| **Name** | Aayukta Singh |
| **Education** | M.Sc (Plant Pathology)<br>B.Sc (Agriculture) |
| **Experience** | 1 year |
| **Field** | Agriculture |
| **Approved Roles** | |

| | | | |
|---|---|---|---|
| **Team Leader** | Yes (VM only) | | |
| **Validator** | Yes (VM only) | | |
| **Verifier** | Yes (VM only) | | |
| **Methodology Expert** | NO | | |
| **Local expert** | NO | | |
| **Financial Expert** | NO | | |
| **Technical Reviewer** | NO | | |
| **TA Expert (X.X)** | NO | | |
| | | | |
| | | | |
| **Reviewed by** | Shifali Guleria (Quality Manager) | **Date** | 04/01/2024 |
| **Approved by** | Deepika Mahala (Technical Manager) | **Date** | 04/01/2024 |

| **Competence Statement** | |
|---|---|
| **Name** | Ashok Gautam |
| **Country** | India |
| **Education** | M. Sc. (Environmental Sciences)<br>M. Tech. (Energy & Environmental Management) |
| **Experience** | 16 Years + |
| **Field** | Energy, Climate Change & Environment |
| **Approved Roles** | |
| **Team Leader** | YES |
| **Validator** | YES |
| **Verifier** | YES |

| | | | |
|---|---|---|---|
| **Methodology Expert** | AMS-I.D., AMS-I.A., AMS-I.C., AMS-I.E, AMS-II.D., AMS-II.G., AMS-III.E., AMS-III.H., AMS-III.Q, AMS-III.Z., AMS-III.AV., AMS III.AR, AM0029, AM0025, AM0056, ACM0001, ACM0002, ACM0004, ACM0012, ACM0006, AM0018, ACM0017, ACM0009, AM0034, AMS.I.B, ACM0016, AMS-III.BL, AMS-II.L, AMS-I.I., AMS-III.A.O., ACM0010, ACM0025 | | |
| **Local expert** | YES (India) | | |
| **Financial Expert** | YES | | |
| **Technical Reviewer** | YES | | |
| **TA Expert** | YES (TA 1.1, TA 1.2, TA 3.1, TA 13.1) | | |
| | | | |
| **Reviewed by** | Shifali Guleria | **Date** | 06/03/2023 |
| **Approved by** | Deepika Mahala | **Date** | 06/03/2023 |

| **Competence Statement** | |
|---|---|
| **Name** | Kalpana Arora |
| **Country** | India |
| **Education** | M.Sc. (MicroBiology), Ph.D. (Waste Management) |
| **Experience** | 5.5 Years + |
| **Field** | Waste management, Animal Waste, Environment |
| **Approved Roles** | |
| **Team Leader** | NO |
| **Validator** | NO |
| **Verifier** | NO |
| **Methodology Expert** | NO |
| **Local Expert** | NO |
| **Financial Expert** | NO |
| **Technical Reviewer** | NO |
| **TA Expert** | YES (TA 13.2 & TA 15.1) |

| | | | |
|---|---|---|---|
| | | | |
| **Reviewed by** | Shreya Garg | **Date** | 17/01/2019 |
| **Approved by** | Anshika Gupta | **Date** | 18/01/2019 |

# APPENDIX III: LIST OF DOCUMENTS REFERRED

| Ref no. | Author | Title | Reference to the document | Provider |
|---|---|---|---|---|
| 1 | PP | VCS Project Description & Monitoring Report (PD&MR) | Version 2.0 dated 01/02/2022 | PP |
| 2 | VCS | VCS PD&MR template | Version 4.2 | Others |
| 3 | VCS | VCS Program Guide | Version 4.2 | Others |
| 4 | VCS | VCS Standard | Version 4.4 | Others |
| 5 | VCS | VCS Validation and Verification Manual | Version 3.2 | Others |
| 6 | VCS | VCS Program Definitions | Version 4.2 | Others |

| | | | | |
|---|---|---|---|---|
| 7 | UNFCCC | A. AMS-III.D., "Methane Recovery in animal manure management systems".<br>B. AMS-I.D., "Grid-connected renewable electricity generation". | Version 21.0<br><br>Version 18.0 | Others |
| 8 | UNFCCC | A. CDM Tool 14 -" Project and leakage emissions from anaerobic digesters" (version 02.0)<br>B. CDM TOOL 06 - "Project Emissions from Flaring" (version 04.0).<br>C. CDM TOOL 05 - "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (version 03.0).<br>D. CDM TOOL 08 – "Tool to determine the mass flow of a greenhouse gas in a gaseous stream." (Version 3.0)<br>E. CDM TOOL 12 - "Project and leakage emissions from transportation of freight." (Version 01.1.0)<br>F. CDM TOOL 03 - "Tool to calculate project or leakage CO2 emissions from fossil fuel combustion" (version 03.0)<br>G. CDM TOOL 07 - "Tool to calculate the emission factor for an electricity system" (version 07.0)<br>H. CDM TOOL 10 - "Tool to determine the remaining lifetime of equipment" (version 01.0)<br>I. CDM TOOL 11 - "Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period." (Version 03.0.1) | - | Others |

| | | | | |
|---|---|---|---|---|
| | | a. | | |
| 9 | UNFCCC | CDM Standard: Sampling and surveys for CDM project activities and programmes of activities | Version 9.0 | Others |
| 10 | UNFCCC | CDM Guideline: Sampling and surveys for CDM project activities and programmes of activities | Version 4.0 | Others |
| 11 | VERRA | VCS Project webpage<br><br>https://registry.verra.org/app/projectDetail/VCS/4025 | Last accessed on 13/06/2023 | Others |
| 12 | PP | Proof of project start date | - | Others |
| 13 | PP | Supportive of start date of Construction | - | PP |
| 14 | PP | Commissioning certificate | - | PP |
| 15 | PP | Proof of Ownership (Yılmazlar Madencilik Sanayi ve Ticaret Limited Şirketi) | - | Others |
| 16 | PP | Agreement with the poultry farms for chicken litter | - | PP |
| 17 | PP | Equipment purchases record | - | PP |
| 18 | PP | Manufacturer/technical specifications of the equipment installed at the project-<br>• Solid Waste Receiving Unit<br>• Liquid Waste Receiving Unit<br>• Material Mixing Unit<br>• Anaerobic Digesters<br>• Separator Units<br>• Gas Engine and Flare Units<br>• Biofilters<br>• Biogas analyzer<br>• Anaerobic lagoons<br>• Gas engine<br>• Gas blowers<br>• Generator<br>• Gas Blower<br>• Transformer<br>• Flaring system<br><br>Electricity Meters | - | PP |
| 19 | PP | Fuel consumption record of DG set. | - | PP |
| 20 | PP | Copy of Mechanical Separation, Bio-Drying and Biomethanization Plants and | - | PP |

| | | | | |
|---|---|---|---|---|
| | | Fermented Product Management Communiqué | | |
| 21 | PP | a. Declaration confirming that the GHG Emission reductions or removals generated by the Project will not be used for compliance with an emission trading program or to meet binding limits on GHG Emissions.<br><br>b. ownership evidence | - | PP |
| 22 | PP | EPIAS electricity generation records. | - | PP |
| 23 | PP | Feasibility Report | - | PP |
| 24 | PP | LSC report | - | PP |
| 25 | PP | Undertaking by the PP, that the project is not registered on any other registry to avoid double counting of the project. | - | PP |
| 26 | PP | Social Security Records | - | PP |
| 27 | PP | Data sheet of Plant load factor since commissioning. | - | PP |
| 28 | PP | Proof of the tariff rate provided by the Turkish state | - | PP |
| 29 | PP | Project Identification Document (PID) submitted to Ministry of Environment, Urbanization and Climate Change | - | PP |
| 30 | PP | Land Purchase Agreement. | - | PP |
| 31 | PP | Power Purchase Agreement | - | PP |
| 32 | PP | Waste collection data | - | PP |
| 33 | PP | Flow meter data | - | PP |
| 34 | PP | On-site data records for substantiating the following parameters –<br>a. Quantity manure brought to the plant<br>b. Average truck capacity for transportation | - | PP |
| 35 | PP | Plant's log for substantiating the following parameters –<br>a. Average animal weight of a defined livestock population at the project site<br>b. Fraction of manure handled in system j in the Project<br>c. Number of animals produced annually of type LT for the year y | - | PP |

| | | | | |
|---|---|---|---|---|
| | | Number of days animal is alive in the farm in the year y | | |
| 36 | PP | Türkiye emission factor datasheet, 2020 | - | PP |
| 37 | PP | Record of captured methane fed to the flare | - | PP |
| 38 | PP | Calibration certificates for electricity meters | - | PP |
| 39 | PP | KML files | - | PP |
| 40 | PP | Emission reduction calculation Sheet | - | PP |
| 41 | PP | Provisional Acceptance Certificate | - | PP |
| 42 | PP | Generation License | - | PP |
| 43 | PP | Connection Agreement | - | PP |
| 44 | PP | SV Records | - | PP |
| 45 | PP | Single line diagram | | PP |
| 46 | PP | EIA exemption certificate | | PP |
| 47 | PP | Project Identification Document | | PP |
| 48 | PP | Feasibility Study | | PP |
| 49 | Others | IPCC 5th Assessment Report | | IPCC |
| 50 | PP | Baseline survey records | | PP |
| 51 | PP | Flaring records | | PP |

# APPENDIX IV: CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS AND FORWARD ACTION REQUESTS

**Table 1. Remaining FAR from validation and/or previous verification**

| **FAR ID** | | **Section no.** | - | **Date :** DD/MM/YYYY |
|---|---|---|---|---|
| **Description of FAR** | | | | |
| *NA* | | | | |
| **Project participant response** | | | | **Date :** DD/MM/YYYY |
| *NA* | | | | |
| **Documentation provided by project participant** | | | | |
| *NA* | | | | |
| **VVB assessment** | | | | **Date:** DD/MM/YYYY |
| NA | | | | |

**Table 2. CL from this verification**

| **CL ID** | 01 | **Section no.** | 3.1 | **Date :** 24/05/2023 |
|---|---|---|---|---|
| **Description of CL** | | | | |
| In section 1.11 of the PD & MR, the PP states that in the mixing unit, to prevent the odor the PP uses biofilters, the biofilters needs to be replaced periodically. The PP is requested to clarify the frequency of how often the biofilters are replaced. | | | | |
| **Project participant response** | | | | **Date :** 01/06/2023 |
| *At the beginning it was planned to put biofilters and it was stated in PID of the project. Then, during the site visit, it has been seen that this idea was given up and instead of it, top of the pools has been closed. Related section has been revised accordingly.* | | | | |
| **Documentation provided by project participant** | | | | |
| | | | | |
| **DOE assessment** | | | | **Date:** 11/06/2023 |
| The biofilters are deemed ineffective, therefore, it has now been replaced with the top of the pools now being covered. Th VVB has reviewed section 1.11 of the joint PD & MR and confirms That the information provided is found to be appropriate.<br><br>Hence, CL#01 now stands closed. | | | | |

| CL ID | 02 | Section no. | 3.1 | Date : 22/05/2023 |
|---|---|---|---|---|
| **Description of CL** | | | | |
| The coordinates provided for the project location are corner co-ordinates, however, there are 20 corner coordinates provided for the project location –<br>1. The relevance of the 20 geo-coordinates is not clear. The PP is requested to clarify.<br>2. The PP is receiving waste from 12 different farms, the PP is requested to provide the geodetic coordinates of the 12 farms as well. | | | | |
| **Project participant response** | | | | **Date :** 01/06/2023 |
| *1- Geo-coordinates were taken from generation license in ED50 UTM format. Now, it has been converted to WGS84, decimal degree format.*<br>*2- Googleearth document (in .kml format) that shows all farms and Yılmazlar BPP has been provided now.* | | | | |
| **Documentation provided by project participant** | | | | |
| | | | | |
| **DOE assessment** | | | | **Date:** 11/06/2023 |
| The PP has now provided the geo-coordinates of the various corners of the plant in WGS84 format, further, the VVB has reviewed the KML file provided by the PP and confirms that the geo-coordinates of the project location were appropriate.<br><br>Hence, CL#02 stands closed. | | | | |

| CL ID | 03 | Section no. | 3.3.2 | Date : 22/05/2023 |
|---|---|---|---|---|
| **Description of CL** | | | | |
| The Local stakeholder consultation of the Project was conducted on 15/11/2022 and the start date of the Project was 27/06/2021. However, In-line to the project standard, para 3.18.3, version 4.4. "The project proponent shall conduct a local stakeholder consultation prior to validation as a way to inform the design of the project and maximize participation from stakeholders." The PP is requested to clarify if there has been any LSC meetings conducted prior to the project start date. If yes, kindly provide evidence to substantiate the same. | | | | |
| **Project participant response** | | | | **Date :** 01/06/2023 |
| *Validation process has started by initial submission of PD & MR to VVB. LSC meeting were held on 15/11/2022 which is earlier than this date. Before this date, there is not any other LSC meeting since the project was exempted from EIA as well.* | | | | |
| **Documentation provided by project participant** | | | | |
| | | | | |
| **DOE assessment** | | | | **Date:** 11/06/2023 |
| The VVB has reviewed the EIA report as well as the LSC records provided, further, since the project is still under validation, the condition is met appropriately.<br><br>Hence, CL#03 stands closed. | | | | |

<table>
<tr><td>CL ID</td><td>04</td><td>Section no.</td><td>3.4.8</td><td>Date : 22/05/2023</td></tr>
<tr><td colspan="5">Description of CL</td></tr>
<tr><td colspan="5">The vehicle used for transportation of waste material is a truck. However, for the parameter, “$EF_{CO2,transport}$” enlisted in section 6.1 of the joint PD & MR, the PP has used the emission factor for a light vehicle and also the tool applied for the source of the value applied is of an older version. The PP is requested to revise the emission factor values and also provide clarification on why the emission factor of light vehicle is applied here.</td></tr>
<tr><td colspan="4">Project participant response</td><td>Date : 01/06/2023</td></tr>
<tr><td colspan="5">First of all, version of the tool has been revised now. In accordance with the tool, light vehicle is defined as the vehicle with gross load capacity less than 26 ton. Each truck bring manure with weight of around 24-26 tons. To get the highest project emission, upper limit of light vehicle is taken and emission factor for light vehicle is taken which is higher. This approach is conservative since the PE is taken highest possible by this way.</td></tr>
<tr><td colspan="5">Documentation provided by project participant</td></tr>
<tr><td colspan="5"></td></tr>
<tr><td colspan="4">DOE assessment</td><td>Date: 11/06/2023</td></tr>
<tr><td colspan="5">The PP has taken a conservative approach with taking the upper limit of light vehicle and emission factor of light vehicle, which is higher, and also since the manure brought in by the truck is around 24-26 tons on daily basis, the vehicle is considered light.<br><br>Hence, CL#04 stands closed.</td></tr>
</table>

Table 3. CAR from this verification

<table>
<tr><td>CAR ID</td><td>01</td><td>Section no.</td><td>3.1</td><td>Date : 22/05/2023</td></tr>
<tr><td colspan="5">Description of CAR</td></tr>
<tr><td colspan="5">The PP has stated the eligibility conditions in-line to version 4.3 of the VCS standard. However, the conditions need to be justified on the basis of latest version of the methodology. The PP is requested to justify the eligibility criteria of the project in-line to para 2.1.1 of the VCS standard, version 4.4.</td></tr>
<tr><td colspan="4">Project participant response</td><td>Date : 01/06/2023</td></tr>
<tr><td colspan="5">Revised according to the VCS standard 4.4.</td></tr>
<tr><td colspan="5">Documentation provided by project participant</td></tr>
<tr><td colspan="5"></td></tr>
<tr><td colspan="4">DOE assessment</td><td>Date: 11/06/2023</td></tr>
<tr><td colspan="5">The PP has now revised the joint PD & MR in-line to the latest applicable VCS standard.<br><br>Hence, CAR#01 stands closed.</td></tr>
</table>

| **CAR ID** | 02 | **Section no.** | 3.1 | **Date :** 24/05/2023 |
|---|---|---|---|---|
| **Description of CL** | | | | |
| The PP is requested to add the SDG targets as well as the indicators relevant to the SDGs the project is targeting in section 17.1 and 17.2 of the PD & MR. | | | | |
| **Project participant response** | | | | **Date :** 01/06/2023 |
| *In part 1.17.1, indicators have been mentioned now. In part 1.17.2. Indicators and related explanations were already mentioned in table 3.* | | | | |
| **Documentation provided by project participant** | | | | |
| | | | | |
| **DOE assessment** | | | | **Date:** 11/06/2023 |
| The PP has now added relevant SDG targets and indicators in the joint PD & MR and it is found to be appropriate.<br>Hence, CAR#02 stands closed. | | | | |

| **CAR ID** | 03 | **Section no.** | 3.1 | **Date :** 24/05/2023 |
|---|---|---|---|---|
| **Description of CL** | | | | |
| In-line to the joint PD & MR template, version 4.2, the PP is requested to provide a brief description of the quantifiable impact of the project's activities related to the SDG indicator, during the monitoring period and over the project lifetime in column "Current project contributions" and "contributions over project lifetime" table 3 of the joint PD & MR. | | | | |
| **Project participant response** | | | | **Date :** 01/06/2023 |
| *Related explanations and brief descriptions have been added under column relevant columns.* | | | | |
| **Documentation provided by project participant** | | | | |
| | | | | |
| **DOE assessment** | | | | **Date:** 11/06/20234 |
| The VVB has reviewed section 1.17.2 of the joint PD & MR, and it has been found that the corrections made are appropriate.<br>Hence, CAR#03 stands closed. | | | | |

| **CAR ID** | 04 | **Section no.** | 3.4 | **Date :** 24/05/2023 |
|---|---|---|---|---|
| **Description of CL** | | | | |
| The PP has mentioned the relevant tools applicable to the methodologies in the joint PD & MR. However, the applicability conditions of the tools are missing from the joint PD & MR. The PP is requested to demonstrate and justify the applicability conditions of the tools applied in the Project in section 3.2 of the Joint PD & MR. | | | | |
| **Project participant response** | | | | **Date :** 01/06/2023 |

| |
|---|
| *Applicability conditions of tools that have been used are added now.* |
| **Documentation provided by project participant** |
| |
| **DOE assessment** — **Date:** 11/06/2023 |
| The section 3.2 of the PD & MR has now been revised with all the added applicability conditions of the tools and the same has also been justified in the joint PD & MR.<br><br>Hence, CAR#04 stands closed. |

| **CAR ID** | 05 | **Section no.** | 3.1 | **Date :** 24/05/2023 |
|---|---|---|---|---|
| **Description of CL** | | | | |
| There have been no comments found on the Verra registry for the project. The PP is requested to revise this section accordingly. | | | | |
| **Project participant response** | | | | **Date :** 01/06/2023 |
| *The section is revised now.* | | | | |
| **Documentation provided by project participant** | | | | |
| | | | | |
| **DOE assessment** | | | | **Date:** 11/06/2023 |
| The section 2.4 of the PD & MR has now been revised.<br><br>Hence, CAR#05 stands closed. | | | | |

| **CAR ID** | 06 | **Section no.** | 3.4.8 | **Date :** 22/05/2023 |
|---|---|---|---|---|
| **Description of CAR** | | | | |
| The following values have been found inconsistent in the joint PD & MR –<br>a. For dairy cattle, the value of VSLT,y is 1816.0575 in the ER sheet. It is found to be inconsistent with the $BE_{Kayı\ Farm,y}$ calculation in section 5.1 of the joint PD & MR. Further, the value of $BE_y$ is also found to be inconsistent with the F41 cell number of “baseline” sheet of ER sheet.<br>b. The value of estimated GHG emissions in the year 27th June 2021 – 31st December 2021 mentioned in section 1.10 of the Joint PD & MR is found to be inconsistent with the cell H5 of the ER sheet.<br>c. Sectoral scope 13 is titled waste handling and disposal, however it is mentioned waste recovery and disposal in section 1.2 of the Joint PD & MR.<br>d. The value for parameter of $VS_{LT,y}$ of dairy cattle mentioned in section is found to be inconsistent from the ER sheet, the PP is requested to revise this. | | | | |
| **Project participant response** | | | | **Date :** 01/06/2023 |

| | |
|---|---|
| *a. VSLT value is corrected in PD & MR. The rest of the calculations depending on this value also corrected now.*<br>*b. Value is corrected now.*<br>*c. Title has been corrected now.*<br>*d. VSLT values for dairy cattle has been corrected in the tables as well.* | |
| **Documentation provided by project participant** | |
| | |
| **DOE assessment** | **Date:** 11/06/2023 |
| The VVB has reviewed the revised PD & MR and confirms that the values found inconsistent from the ER sheet have now been corrected. Further, the inconsistency related to the PD & MR has also been corrected.<br><br>Hence, CAR#06 stands closed. | |

| **CAR ID** | 07 | **Section no.** | 4.1 | **Date :** 22/05/2023 |
|---|---|---|---|---|
| **Description of CAR** | | | | |
| The estimated baseline emissions summary for AMS-III.D mentioned in section 5.4 of the joint PD & MR is found to be inconsistent from the tab "ER Summary" of the ER sheet. The PP is requested to revise this. | | | | |
| **Project participant response** | | | | **Date :** 01/06/2023 |
| *The value is corrected now.* | | | | |
| **Documentation provided by project participant** | | | | |
| | | | | |
| **DOE assessment** | | | | **Date:** 11/06/2023 |
| The VVB has reviewed the revised PD & MR and confirms that the PP has now corrected the values of estimated baseline emissions and it is now found to be consistent with the ER sheet.<br><br>Hence, CAR#07 stands closed. | | | | |